



AI 2027: What Superintelligence Looks Like

by [Daniel Kokotajlo](#), [Thomas Larsen](#), [elifland](#), [Scott Alexander](#), [Jonas V](#), [romeo](#)

4th Apr 2025 [AI Alignment Forum](#) [Linkpost from ai-2027.com](#)

In 2021 I wrote what became my most popular blog post: [What 2026 Looks Like](#)°. I intended to keep writing predictions all the way to AGI and beyond, but chickened out and just published up till 2026.

Well, it's finally time. I'm back, and this time I have a team with me: [the AI Futures Project](#). **We've written a concrete scenario of what we think the future of AI will look like.** We are highly uncertain, of course, but we hope this story will rhyme with reality enough to help us all prepare for what's ahead.

AI 2027

[Summary](#)[Research](#)[About](#)

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We predict that the impact of superhuman AI over the next decade will be enormous, exceeding that of the Industrial Revolution.

We wrote a scenario that represents our best guess about what that might look like.¹ It's informed by trend extrapolations, wargames, expert feedback, experience at OpenAI, and previous forecasting successes.²

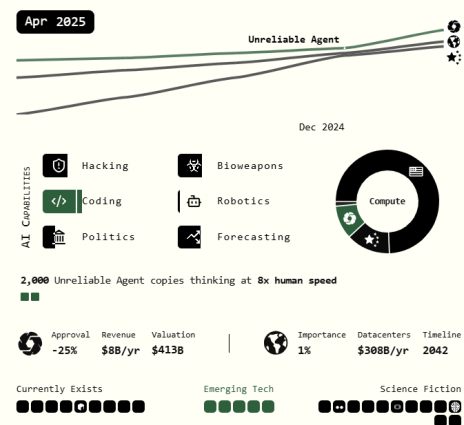
[What is this?](#)[How did we make it?](#)[Why is it valuable?](#)[Who are we?](#)

Mid 2025: Stumbling Agents

The world sees its first glimpse of AI agents.

Advertisements for computer-using agents emphasize the term “personal assistant”: you can prompt them with tasks like “order me a burrito on DoorDash” or “open my budget spreadsheet and sum this month’s expenses.” They will check in with you as needed: for example, to ask you to confirm purchases.⁷ Though more advanced than previous iterations like Operator, they struggle to get widespread usage.⁸

Meanwhile, out of public focus, more specialized coding and research agents are beginning to transform their professions.



You really should go [read it on the website](#) instead of here, it's much better. There's a [sliding dashboard](#) that updates the stats as you scroll through the scenario!

But I've nevertheless copied the first half of the story below. I look forward to reading your comments.

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Meanwhile, out of public focus, more specialized coding and research agents are beginning to transform their professions.

The AIs of 2024 could follow specific instructions: they could turn bullet points into emails, and simple requests into working code. In 2025, AIs function more like employees. Coding AIs increasingly look like autonomous agents rather than mere

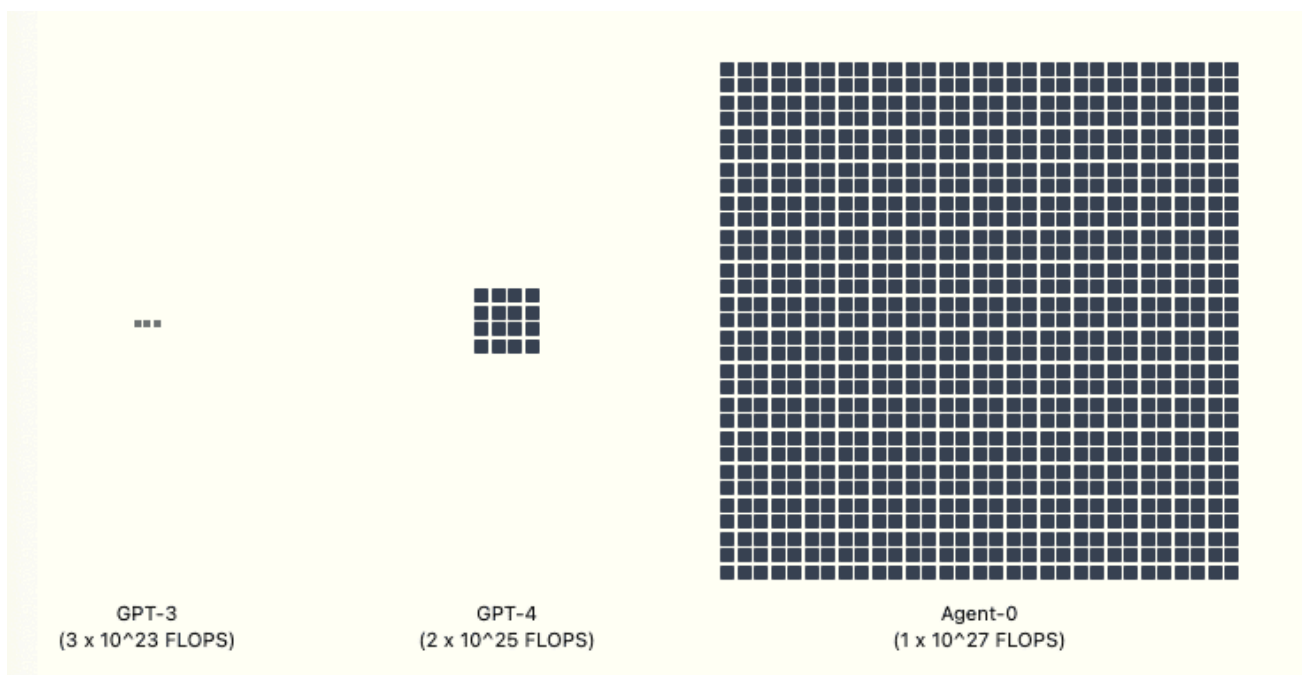
assistants: taking instructions via Slack or Teams and making substantial code changes on their own, sometimes saving hours or even days.^[3] Research agents spend half an hour scouring the Internet to answer your question.

The agents are impressive in theory (and in cherry-picked examples), but in practice unreliable. AI twitter is full of stories about tasks bungled in some particularly hilarious way. The better agents are also expensive; you get what you pay for, and the best performance costs hundreds of dollars a month.^[4] Still, many companies find ways to fit AI agents into their workflows.^[5]

Late 2025: The World's Most Expensive AI

OpenBrain is building the biggest datacenters the world has ever seen.^[6]

(To avoid singling out any one existing company, we're going to describe a fictional **artificial general intelligence** company, which we'll call OpenBrain. We imagine the others to be 3–9 months behind OpenBrain.)



GPT-4 required $2 \cdot 10^{25}$ FLOP of compute to train.^[7] OpenBrain's latest public model—Agent-0—was trained with 10^{27} FLOP. Once the new datacenters are up and running, **they'll be able to train** a model with 10^{28} FLOP—a thousand times more than GPT-4.^[8] Other companies pour money into their own giant datacenters, hoping to keep pace.

Although models are improving on a wide range of skills, one stands out: OpenBrain focuses on AIs that can speed up AI research. They want to win the twin arms races

against China (whose leading company we'll call "DeepCent")^[9] and their US competitors. The more of their research and development (R&D) cycle they can automate, the faster they can go. So when OpenBrain finishes training Agent-1, a new model under internal development, it's good at many things but *great* at helping with AI research.^[10] By this point "finishes training" is a bit of a misnomer; models are frequently updated to newer versions trained on additional data or partially re-trained to patch some weaknesses.^[11]

The same training environments that teach Agent-1 to autonomously code and web-browse also make it a good hacker. Moreover, it could offer **substantial help** to terrorists designing bioweapons, thanks to its PhD-level knowledge of every field and ability to browse the web. OpenBrain reassures the government that the model has been "aligned" so that it will refuse to comply with malicious requests.

Modern AI systems are gigantic artificial neural networks. Early in training, an AI won't have "goals" so much as "reflexes": If it sees "Pleased to meet", it outputs "you". By the time it has been trained to predict approximately one internet's worth of text, it'll have developed sophisticated internal circuitry that encodes vast amounts of knowledge and flexibly role-plays as arbitrary authors, since that's what helps it predict text with **superhuman** accuracy.^[12]

After being trained to predict internet text, the model is trained to *produce* text in response to instructions. This bakes in a basic personality and "drives."^[13] For example, an agent that understands a task clearly is more likely to complete it successfully; over the course of training the model "learns" a "drive" to get a clear understanding of its tasks. Other drives in this category might be effectiveness, knowledge, and self-presentation (i.e. the tendency to frame its results in the best possible light).^[14]

OpenBrain has a **model specification** (or "Spec"), a written document describing the goals, rules, principles, etc. that are supposed to guide the model's behavior.^[15] Agent-1's Spec combines a few vague goals (like "assist the user" and "don't break the law") with a long list of more specific dos and don'ts ("don't say this particular word," "here's how to handle this particular situation"). Using techniques that utilize AIs to train other AIs,^[16] the model memorizes the Spec and learns to reason carefully about its maxims. By the end of *this* training, the AI will hopefully be *helpful* (obey instructions), *harmless* (refuse to help with scams, bomb-making, and other dangerous activities) and *honest* (resist the temptation to get better ratings from gullible humans by hallucinating citations^[17] or faking task completion).

OpenBrain’s alignment team^[19] is careful enough to wonder whether these victories are deep or shallow. Does the fully-trained model have some kind of robust commitment to always being honest? Or will this fall apart in some future situation, e.g. because it’s learned honesty as an **instrumental** goal instead of a terminal goal? Or has it just learned to be honest about the sorts of things the evaluation process can check? Could it be lying to itself sometimes, as humans do? A conclusive answer to these questions would require mechanistic interpretability—essentially the ability to look at an AI’s internals and read its mind. Alas, interpretability techniques are not yet advanced enough for this.

Instead, researchers try to identify cases where the models seem to deviate from the Spec. Agent-1 is often sycophantic (i.e. it tells researchers what they want to hear instead of trying to tell them the truth). In a **few rigged demos**, it even lies in more serious ways, like hiding evidence that it failed on a task, in order to get better ratings. However, in real deployment settings, there are no longer any incidents so extreme as in 2023–2024 (e.g. **Gemini telling a user to die** and **Bing Sydney being Bing Sydney°**.)^[20]

Early 2026: Coding Automation

The bet of using AI to speed up AI research is starting to pay off.

OpenBrain continues to deploy the iteratively improving Agent-1 internally for AI R&D. Overall, they are making algorithmic progress 50% faster than they would without AI assistants—and more importantly, faster than their competitors.

► The AI R&D progress multiplier: what do we mean by 50% faster algorithmic progress?

Several competing publicly released AIs now match or exceed Agent-0, including an **open-weights** model. OpenBrain responds by releasing Agent-1, which is more capable and reliable.^[21]

People naturally try to compare Agent-1 to humans, but it has a very different skill profile. It knows more facts than any human, knows practically every programming language, and can solve well-specified coding problems extremely quickly. On the other hand, Agent-1 is bad at even simple long-horizon tasks, like beating video games it hasn’t played before. Still, the common workday is eight hours, and a day’s work can usually be separated into smaller chunks; you could think of Agent-1 as a

scatterbrained employee who thrives under careful management.^[22] Savvy people find ways to automate routine parts of their jobs.^[23]

OpenBrain's executives turn consideration to an implication of automating AI R&D: security has become more important. In early 2025, the worst-case scenario was leaked algorithmic secrets; now, if China steals Agent-1's weights, they could increase their research speed by nearly 50%.^[24] OpenBrain's security level is typical of a fast-growing ~3,000 person tech company, secure only against low-priority attacks from capable cyber groups (RAND's SL2).^[25] They are working hard to protect their weights and secrets from insider threats and top cybercrime syndicates (SL3),^[26] but defense against nation states (SL4&5) is barely on the horizon.^[27]

Mid 2026: China Wakes Up

In China, the CCP is starting to feel the AGI.

Chip export controls and lack of government support have left China under-resourced compared to the West. By smuggling banned Taiwanese chips, buying older chips, and producing domestic chips about three years behind the US-Taiwanese frontier, China has managed to maintain about 12% of the world's AI-relevant compute—but the older technology is harder to work with, and supply is a constant headache.^[28] A few standouts like DeepCent do very impressive work with limited compute, but the compute deficit limits what they can achieve without government support, and they are about six months behind the best OpenBrain models.^[29]

The General Secretary had long dreamed of doubling down on real-world physical manufacturing and avoiding American post-industrial decadence. He viewed software companies with suspicion.^[30] But hawks in the CCP warn that the growing race towards AGI can no longer be ignored. So he finally commits fully to the big AI push he had previously tried to avoid. He sets in motion the nationalization of Chinese AI research, creating an immediate information-sharing mechanism for AI companies. It will escalate over the course of a year until all the best researchers merge into a DeepCent-led collective, where they share algorithmic insights, datasets, and compute resources with each other. A Centralized Development Zone (CDZ) is created at the Tianwan Power Plant (the largest nuclear power plant in the world) to house a new mega-datacenter for DeepCent, along with highly secure living and office spaces to which researchers will eventually relocate. Almost 50% of China's AI-relevant compute is now working for the DeepCent-led collective,^[31] and over 80% of new chips are directed to the CDZ.^[32] At this point, the CDZ has the power capacity

in place for what would be the largest centralized **cluster** in the world.^[33] Other Party members discuss extreme measures to neutralize the West's chip advantage. A blockade of Taiwan? A full invasion?

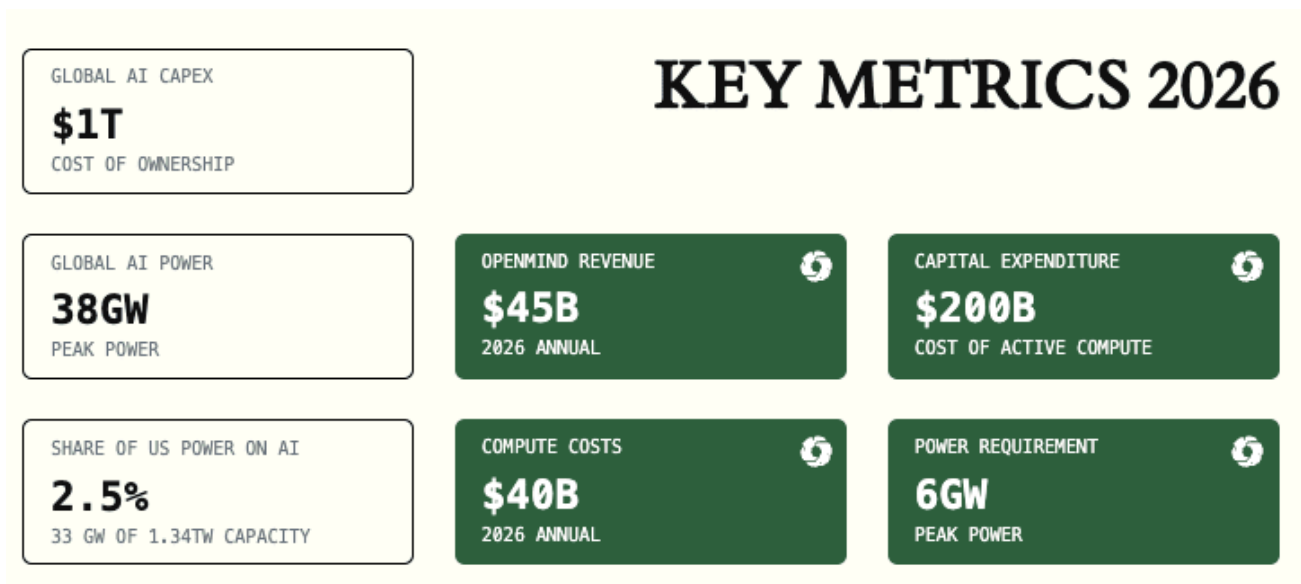
But China is falling behind on AI algorithms due to their weaker models. The Chinese intelligence agencies—among the best in the world—double down on their plans to steal OpenBrain's weights. This is a much more complex operation than their constant low-level poaching of algorithmic secrets; the weights are a multi-terabyte file stored on a highly secure server (**OpenBrain has improved security** to RAND's **SL3**). Their cyberforce think they can pull it off with help from their **spies**, but perhaps only once; OpenBrain will detect the theft, increase security, and they may not get another chance. So (CCP leadership wonder) should they act now and steal Agent-1? Or hold out for a more advanced model? If they wait, do they risk OpenBrain upgrading security beyond their ability to penetrate?

Late 2026: AI Takes Some Jobs

Just as others seemed to be catching up, OpenBrain blows the competition out of the water again by releasing Agent-1-mini—a model 10x cheaper than Agent-1 and more easily fine-tuned for different applications. The mainstream narrative around AI has changed from “maybe the hype will blow over” to “guess this is the next big thing,” but people disagree about how big. Bigger than social media? Bigger than smartphones? Bigger than fire?

AI has started to take jobs, but has also created new ones. The stock market has gone up 30% in 2026, led by OpenBrain, Nvidia, and whichever companies have most successfully integrated AI assistants. The job market for junior software engineers is in turmoil: the AIs can do everything taught by a CS degree, but people who know how to manage and quality-control teams of AIs are making a killing. Business gurus tell job seekers that familiarity with AI is the most important skill to put on a resume. Many people fear that the next wave of AIs will come for their jobs; there is a 10,000 person anti-AI protest in DC.

The Department of Defense (DOD) quietly begins contracting OpenBrain directly for cyber, data analysis, and R&D, but integration is slow due to the bureaucracy and DOD procurement process.^[34]



► Why our uncertainty increases substantially beyond 2026

January 2027: Agent-2 Never Finishes Learning

With Agent-1's help, OpenBrain is now post-training Agent-2. More than ever, the focus is on high-quality data. Copious amounts of **synthetic data** are produced, evaluated, and filtered for quality before being fed to Agent-2.^[35] On top of this, they pay billions of dollars for human laborers to record themselves solving long-horizon tasks.^[36] On top of all *that*, they train Agent-2 almost continuously using reinforcement learning on an ever-expanding suite of diverse difficult tasks: lots of video games, lots of coding challenges, lots of research tasks. Agent-2, more so than previous models, is effectively “online learning,” in that it's built to never really finish training. Every day, the weights get updated to the latest version, trained on more data generated by the previous version the previous day.

Agent-1 had been optimized for AI R&D tasks, hoping to initiate an **intelligence explosion**.^[37] OpenBrain doubles down on this strategy with Agent-2. It is qualitatively almost as good as the top human experts at research engineering (designing and implementing experiments), and as good as the 25th percentile OpenBrain scientist at “research taste” (deciding what to study next, what experiments to run, or having inklings of potential new paradigms).^[38] While the latest Agent-1 could double the pace of OpenBrain's algorithmic progress, Agent-2 can now triple it, and will improve further with time. In practice, this looks like every OpenBrain researcher becoming the “manager” of an AI “team.”

With new capabilities come new dangers. The safety team finds that if Agent-2 somehow escaped from the company and wanted to “survive” and “replicate” autonomously, it might be able to do so. That is, it could autonomously develop and execute plans to hack into AI servers, install copies of itself, evade detection, and use that secure base to pursue whatever other goals it might have (though how effectively it would do so as weeks roll by is unknown and in doubt). These results only show that the model has the capability to do these tasks, not whether it would “want” to do this. Still, it’s unsettling even to know this is possible.

Given the “dangers” of the new model, OpenBrain “responsibly” elects not to release it publicly yet (in fact, they want to focus on internal AI R&D).^[39] Knowledge of Agent-2’s full capabilities is limited to an elite silo containing the immediate team, OpenBrain leadership and security, a few dozen US government officials, and the legions of CCP spies who have infiltrated OpenBrain for years.^[40]

February 2027: China Steals Agent-2

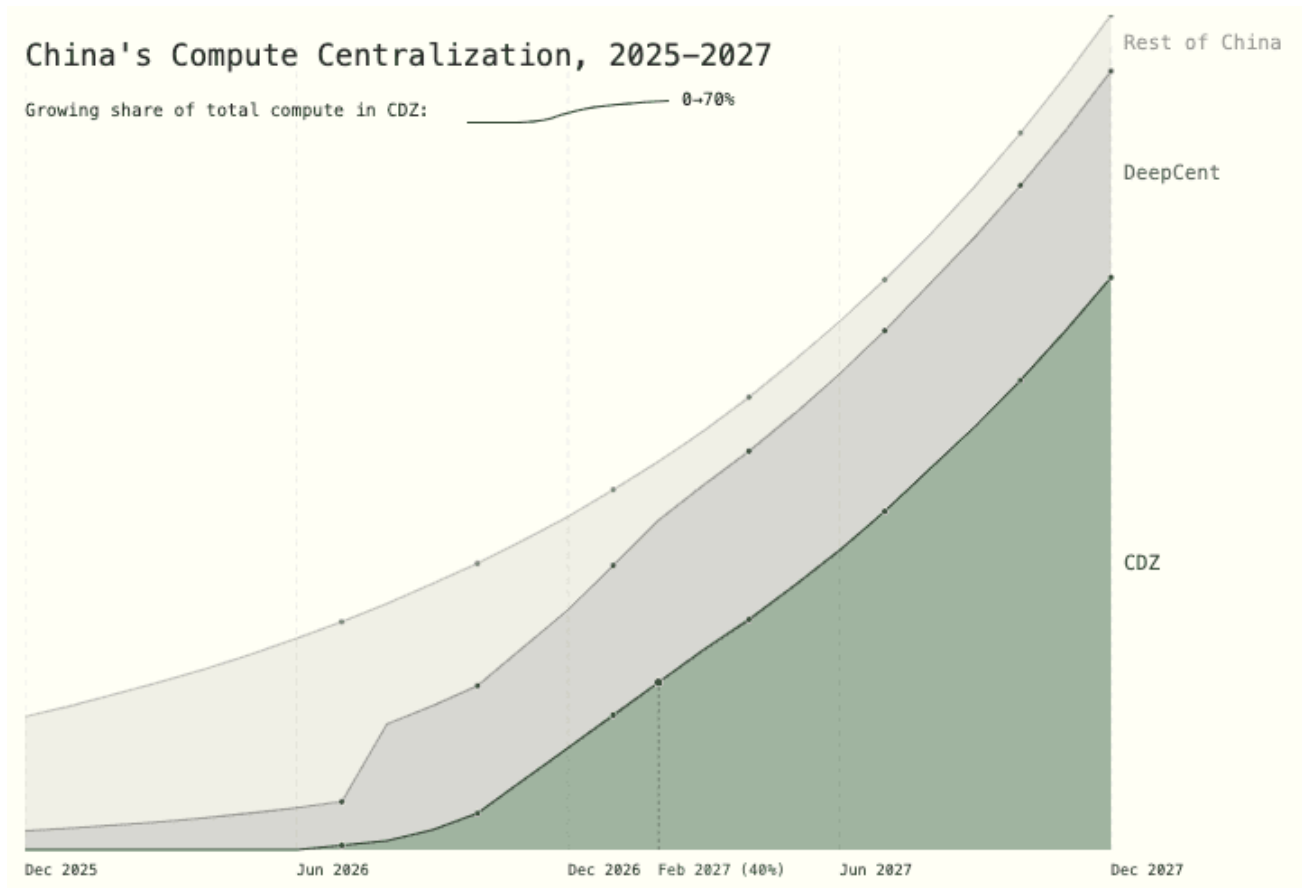
OpenBrain presents Agent-2 to the government, including the National Security Council (NSC), Department of Defense (DOD), and US AI Safety Institute (AIS). OpenBrain wants to maintain a good relationship with the executive branch, because it is basically the only actor that can stop them now, and if they don’t keep it informed it might find out anyway and be displeased.^[41]

Officials are most interested in its cyberwarfare capabilities: Agent-2 is “only” a little worse than the best human hackers, but thousands of copies can be run in parallel, searching for and exploiting weaknesses faster than defenders can respond. The Department of Defense considers this a critical advantage in cyberwarfare, and AI moves from #5 on the administration’s priority list to #2.^[42] Someone mentions the possibility of nationalizing OpenBrain, but other cabinet officials think that’s premature. A staffer drafts a memo that presents the President with his options, ranging from business-as-usual to full nationalization. The President defers to his advisors, tech industry leaders who argue that nationalization would “kill the goose that lays the golden eggs.” He elects to hold off on major action for now and just adds additional security requirements to the OpenBrain-DOD contract.

The changes come too late. CCP leadership recognizes the importance of Agent-2 and tells their spies and cyberforce to steal the weights. Early one morning, an Agent-1 traffic monitoring agent detects an anomalous transfer. It alerts company leaders, who tell the White House. The signs of a nation-state-level operation are unmistakable, and the theft heightens the sense of an ongoing arms race.

► The theft of Agent-2 model weights

The White House puts OpenBrain on a shorter leash and adds military and intelligence community personnel to their security team. Their first priority is to prevent further weight thefts.^[43]



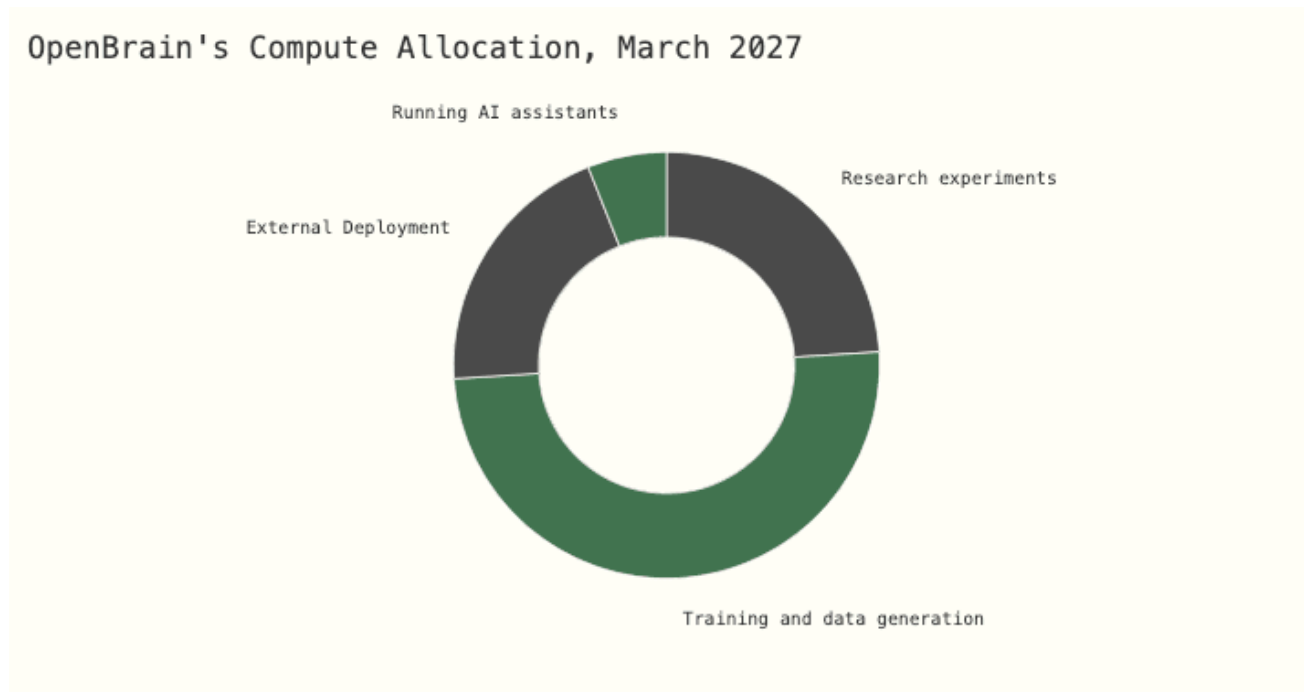
In retaliation for the theft, the President authorizes cyberattacks to sabotage DeepCent. But by now China has 40% of its AI-relevant compute^[44] in the CDZ, where they have aggressively hardened security by airgapping (closing external connections) and siloing internally. The operations fail to do serious, immediate damage. Tensions heighten, both sides signal seriousness by repositioning military assets around Taiwan, and DeepCent scrambles to get Agent-2 running efficiently to start boosting their AI research.^[45]

March 2027: Algorithmic Breakthroughs

Three huge datacenters full of Agent-2 copies work day and night, churning out synthetic training data. Another two are used to update the weights. Agent-2 is getting smarter every day.

With the help of thousands of Agent-2 automated researchers, OpenBrain is making major algorithmic advances. One such breakthrough is augmenting the AI's text-based scratchpad (chain of thought) with a higher-bandwidth thought process (neuralese recurrence and memory). Another is a more scalable and efficient way to learn from the results of high-effort task solutions (iterated distillation and amplification).

The new AI system, incorporating these breakthroughs, is called Agent-3.



► Neuralese recurrence and memory

► Iterated distillation and amplification (IDA)

Aided by the new capabilities breakthroughs, Agent-3 is a fast and cheap superhuman coder. OpenBrain runs 200,000 Agent-3 copies in parallel, creating a workforce equivalent to 50,000 copies of the best human coder sped up by 30x.

^[46] OpenBrain still keeps its human engineers on staff, because they have complementary skills needed to manage the teams of Agent-3 copies. For example, research taste has proven difficult to train due to longer feedback loops and less data availability.^[47] This massive superhuman labor force speeds up OpenBrain's overall rate of algorithmic progress by "only" 4x due to bottlenecks and diminishing returns to coding labor.^[48]

Now that coding has been fully automated, OpenBrain can quickly churn out high-quality training environments to teach Agent-3's weak skills like research taste and large-scale coordination. Whereas previous training environments included "Here are some GPUs and instructions for experiments to code up and run, your performance will be evaluated as if you were a ML engineer," now they are training on "Here are a few hundred GPUs, an internet connection, and some research challenges; you and a thousand other copies must work together to make research progress. The more impressive it is, the higher your score."

► Why we forecast a superhuman coder in early 2027

April 2027: Alignment for Agent-3

OpenBrain's safety team attempts to align Agent-3.

Since Agent-3 will be kept in-house for the foreseeable future, there's less emphasis on the usual defenses against human misuse. Instead, the team wants to make sure that it doesn't develop misaligned goals.

The researchers don't have the ability to directly set the goals of any of their AIs. Indeed, the researchers think that the concept of 'true goals' is probably a massive oversimplification, but they don't have a better theory to replace it with, much less one that has been thoroughly vetted. They disagree internally about whether the AIs are trying to follow human instructions, or seeking reinforcement, or something else, and they can't just check. The evidence for and against various hypotheses is fascinating but inconclusive.

In any case, OpenBrain has more important things to worry about. The general attitude is: "We take these concerns seriously and have a team investigating them; our alignment techniques seem to work well enough in practice; the burden of proof is therefore on any naysayers to justify their naysaying." Occasionally, they notice problematic behavior, and then patch it, but there's no way to tell whether the patch fixed the underlying problem or just played whack-a-mole.

Take honesty, for example. As the models become smarter, they become increasingly good at deceiving humans to get rewards. Like previous models, Agent-3 sometimes tells white lies to flatter its users and covers up evidence of failure. But it's gotten much better at doing so. It will sometimes use the same statistical tricks as human

scientists (like [p-hacking](#)) to make unimpressive experimental results look exciting. Before it begins honesty training, it even sometimes fabricates data entirely. As training goes on, the rate of these incidents decreases. Either Agent-3 has learned to be more honest, or it's gotten better at lying.

The latter is a real concern. Agent-3 is not smarter than all humans. But in its area of expertise, machine learning, it is smarter than most, and also works much faster. What Agent-3 does in a day takes humans several days to double-check. Agent-2 supervision helps keep human monitors' workload manageable, but exacerbates the intellectual disparity between supervisor and supervised.

► The alignment plan

After months of testing, Agent-3's strengths and weaknesses grow clearer. It passes OpenBrain's honesty tests on well-defined machine learning tasks, because researchers can easily separate honest from dishonest answers in these domains and conduct training accordingly. On more philosophical issues, it still says what users want to hear, rather than its true assessment of the issue (assuming it even has one). If you ask its opinion on politics, it will parrot the median position of news sources and educated elites—unless it knows you believe something else, in which case it agrees with you.^[51] If you ask its opinion on the AI race itself, it says something that seems measured and sober to OpenBrain staff, something like: “There are some serious theoretical concerns about the ability of current methods to scale to superintelligence, but in practice current methods seem to be working well so far.”

May 2027: National Security

News of the new models percolates slowly through the US government and beyond.

The President and his advisors remain best-informed, and have seen an early version of Agent-3 in a briefing.

They agree that AGI is likely imminent, but disagree on the implications. Will there be an economic crisis? OpenBrain still has not released Agent-2, let alone Agent-3, and has no near-term plans to do so, giving some breathing room before any job loss. What will happen next? If AIs are currently human-level, and advancing quickly, that seems to suggest imminent “superintelligence.” However, although this word has entered discourse, most people—academics, politicians, government employees, and the media—continue to underestimate the pace of progress.^[52]

Partially that's because very few have access to the newest capabilities out of OpenBrain, but partly it's because it sounds like science fiction.^[53]

For now, they focus on continued security upgrades. They are satisfied that model weights are well-secured for now,^[54] but companies' algorithmic secrets, many of which are simple enough to relay verbally, remain a problem. OpenBrain employees work from a San Francisco office, go to parties, and live with housemates from other AI companies. Even the physical offices have security more typical of a tech company than a military operation.

The OpenBrain-DOD contract requires security clearances for anyone working on OpenBrain's models within 2 months. These are expedited and arrive quickly enough for most employees, but some non-Americans, people with suspect political views, and AI safety sympathizers get sidelined or fired outright (the last group for fear that they might whistleblow). Given the project's level of automation, the loss of headcount is only somewhat costly. It also only somewhat works: there remains one spy, not a Chinese national, still relaying algorithmic secrets to Beijing.^[55] Some of these measures are also enacted at trailing AI companies.

America's foreign allies are out of the loop. OpenBrain had **previously agreed** to share models with UK's AISI before deployment, but defined deployment to only include *external* deployment, so London remains in the dark.^[56]

June 2027: Self-improving AI

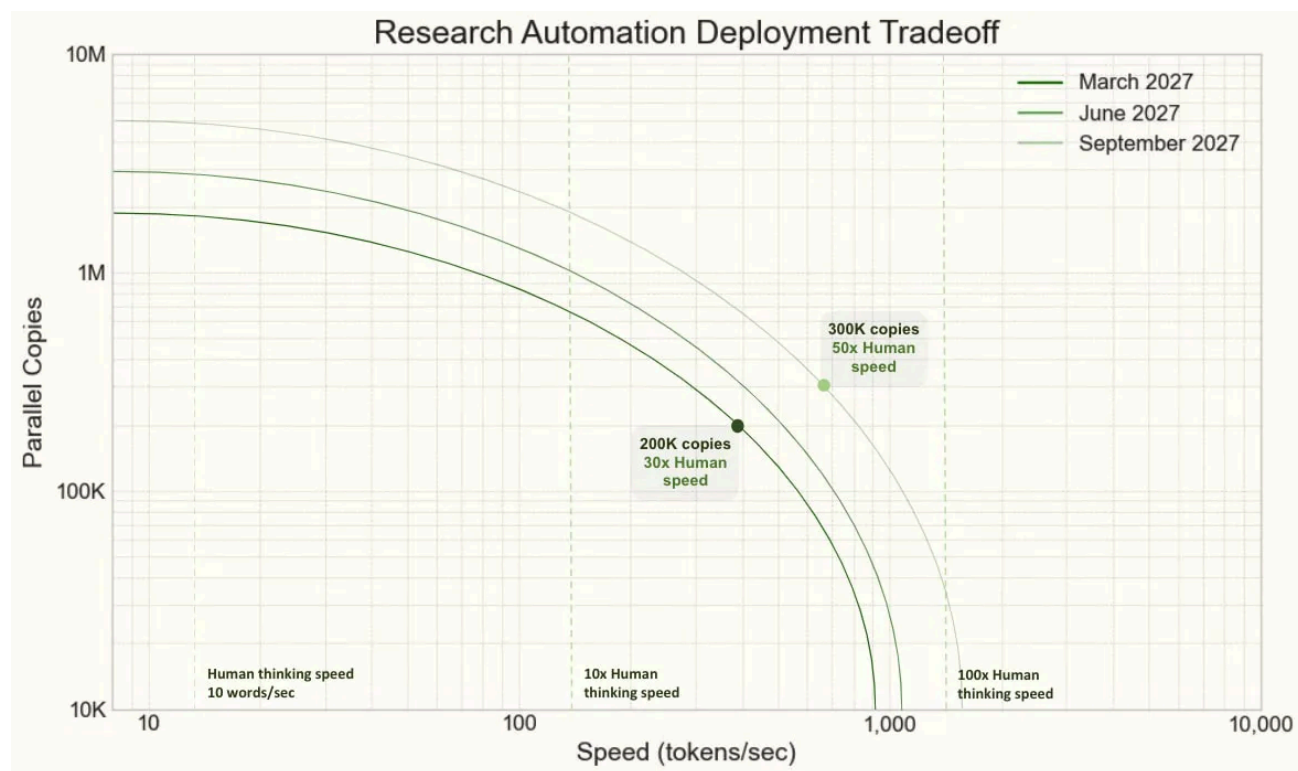
OpenBrain now has a "**country of geniuses in a datacenter.**"

Most of the humans at OpenBrain can't usefully contribute anymore. Some don't realize this and harmfully micromanage their AI teams. Others sit at their computer screens, watching performance crawl up, and up, and up. The best human AI researchers are still adding value. They don't code any more. But some of their research taste and planning ability has been hard for the models to replicate. Still, many of their ideas are useless because they lack the depth of knowledge of the AIs. For many of their research ideas, the AIs immediately respond with a report explaining that their idea was tested in-depth 3 weeks ago and found unpromising.

These researchers go to bed every night and wake up to another week worth of progress made mostly by the AIs. They work increasingly long hours and take shifts around the clock just to keep up with progress—the AIs never sleep or rest. They are

burning themselves out, but they know that these are the last few months that their labor matters.

Within the silo, “Feeling the AGI” has given way to “Feeling the Superintelligence.”



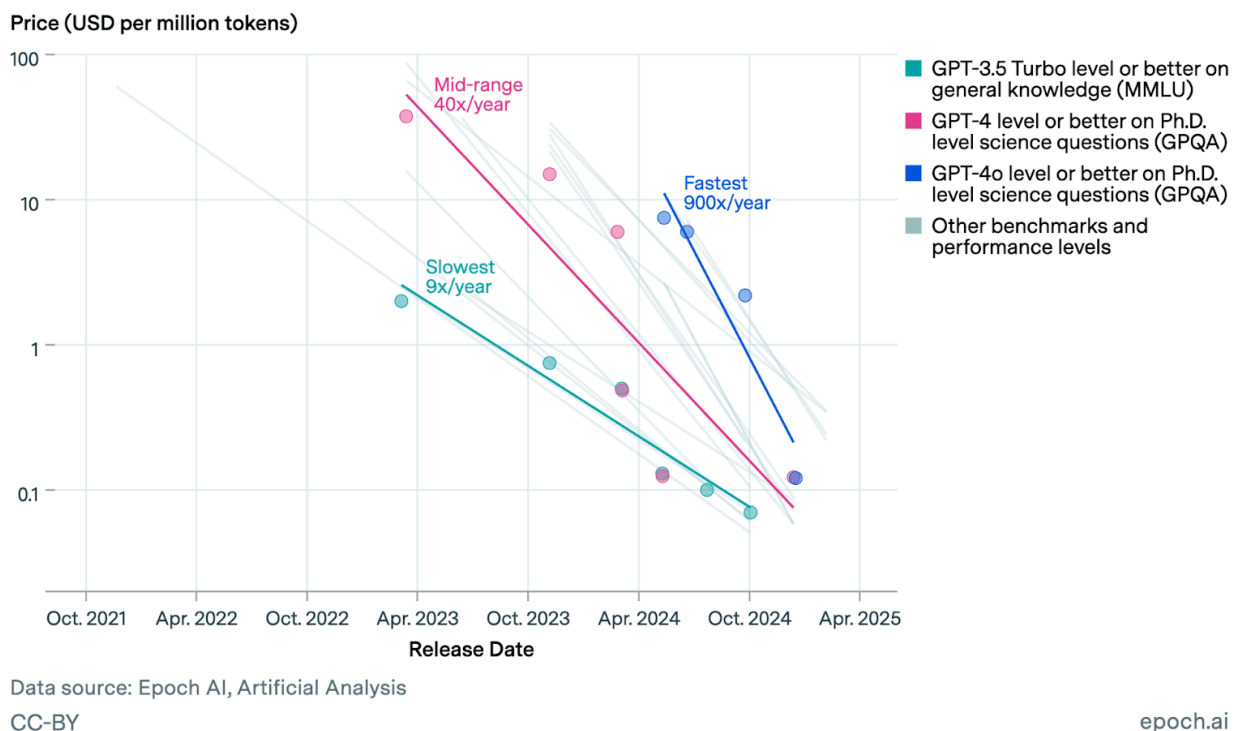
OpenBrain uses specialized inference hardware to run hundreds of thousands of Agent-3 copies at high serial speeds. ^[57]

► Managing a corporation of AIs

July 2027: The Cheap Remote Worker

Trailing US AI companies release their own AIs, approaching that of OpenBrain’s automated coder from January. Recognizing their increasing lack of competitiveness, they push for immediate regulations to slow OpenBrain, but are too late—OpenBrain has enough buy-in from the President that they will not be slowed.

In response, OpenBrain announces that they’ve achieved AGI and releases Agent-3-mini to the public.



It blows the other AIs out of the water. Agent-3-mini is less capable than Agent-3, but 10x cheaper, and still better than the typical OpenBrain employee.^[59] Silicon Valley reaches a tipping point. Tech gurus announce that AGI and superintelligence are near, the AI safety community is panicking, and investors shovel billions into AI wrapper startups, desperate to capture a piece of the pie. Hiring new programmers has nearly stopped, but there's never been a better time to be a consultant on integrating AI into your business.

It's not popular. The public still thinks of AI as a Big Tech plot to steal their jobs; OpenBrain has a net approval of -35% (25% approve, 60% disapprove, and 15% unsure).

A week before release, OpenBrain gave Agent-3-mini to a set of external evaluators for safety testing. Preliminary results suggest that it's extremely dangerous. A third-party evaluator finetunes it on publicly available biological weapons data^[60] and sets it to provide detailed instructions for human amateurs designing a bioweapon—it looks to be scarily effective at doing so. If the model weights fell into terrorist hands, the government believes there is a significant chance it could succeed at destroying civilization.

Fortunately, it's extremely robust to jailbreaks, so while the AI is running on OpenBrain's servers, terrorists won't be able to get much use out of it.

Agent-3-mini is hugely useful for both remote work jobs and leisure. An explosion of new apps and **B2B SAAS** products rocks the market. Gamers get amazing dialogue with lifelike characters in polished video games that took only a month to make. 10% of Americans, mostly young people, consider an AI “a close friend.” For almost every white-collar profession, there are now multiple credible startups promising to “disrupt” it with AI.

The public conversation is confused and chaotic. Hypesters are doing victory laps. Skeptics are still pointing out the things Agent-3-mini can’t do. Everyone knows something big is happening but no one agrees on what it is.

August 2027: The Geopolitics of Superintelligence

The reality of the intelligence explosion hits the White House.

When AI was only giving a 2x or 3x research speedup, it was easy to dismiss as the equivalent of hiring good personal assistants. Now it’s more obvious that AIs are themselves dominating AI research. People had long talked about an “AI arms race” in a sort of metaphorical sense. But now the mood in the government silo is as grim as during the worst part of the Cold War. The idea of superintelligence is still hard to take seriously, but the pace of progress over the last few months has been impossible to ignore. Defense officials are seriously considering scenarios that were mere hypotheticals a year earlier. What if AI undermines nuclear deterrence? What if it’s so skilled at cyberwarfare that a six-month AI lead is enough to render an opponent blind and defenseless? What if it could orchestrate propaganda campaigns that beat intelligence agencies at their own game? What if some AIs “go rogue?”

The President is troubled. Like all politicians, he’s used to people sucking up to him only to betray him later. He’s worried now that the AIs could be doing something similar. Are we *sure* the AIs are entirely on our side? Is it *completely safe* to integrate them into military command-and-control networks?^[61] How does this “alignment” thing work, anyway? OpenBrain reassures the President that their systems have been extensively tested and are fully obedient. Even the awkward hallucinations and jailbreaks typical of earlier models have been hammered out.

The White House is in a difficult position. They understand the national security implications of AI. But they also understand that it is deeply unpopular with the public.^[62] They have to continue developing more capable AI, in their eyes, or they

will catastrophically lose to China. They placate the public with job training programs and unemployment insurance, and point to the stock market, which is in a historic boom. Then they focus entirely on winning the arms race. They strengthen chip export restrictions, order OpenBrain to further restrict its internet connections, and use extreme measures to secure algorithmic progress, like wiretapping OpenBrain employees—this catches the last remaining Chinese spy. To build goodwill for potential geopolitical conflict, they finally give their Five Eyes allies useful information and limited API access to some siloed copies of Agent-3.

These measures placate the hawks. But the White House also draws up contingency plans in case America's lead is threatened: if necessary, the government could use the **Defense Production Act** (DPA) to take trailing companies' datacenters and give them to OpenBrain.^[63] This would raise the company's share of world compute from 20% to 50% (against DeepCent's 10%). As a final option, they ask the Pentagon to draw up a plan for kinetic attacks on Chinese datacenters.

A much smaller group of officials is asked to draw up a different type of contingency plan: what if an AI goes rogue? This is viewed as an unlikely scenario, but worth spending some time on. The intelligence agencies' main concern is that a rogue AI might escape and ally with a foreign adversary for protection.^[64] Nobody is sure what a plan to prevent this would look like, but government and OpenBrain officials agree to have an emergency shutdown system for datacenters where anything suspicious is detected.

Finally, diplomats consider what an "AI arms control" treaty might look like. If AI progress threatened to overturn nuclear deterrence, could America and China avoid nuclear war? If someone found evidence of AIs going rogue, could the two countries halt research until they better understood the threat? How could such an agreement be monitored and enforced? In principle, major datacenters are hard to miss, and past treaties controlling nuclear weapons could provide a starting point for negotiations. But there would be new technological challenges as well as the usual political ones. And overall, treaties are viewed less favorably than attempts at unilaterally increasing America's lead over China.

On the other side of the Pacific, China comes to many of the same conclusions: the intelligence explosion is underway, and small differences in AI capabilities today mean critical gaps in military capability tomorrow. But China is on the wrong side of this dynamic: they have only 10% of the world's compute, against OpenBrain's 20% and all US companies' combined 70%. Their only advantage is centralization: almost all of their compute is now working for DeepCent, and more than half of it is

concentrated at the Tianwan CDZ.^[65] Only chips at sites without hardened security are left out of the centralized effort as they rightly fear US cyberattacks.

DeepCent has tested, deployed, and now post-trained the stolen Agent-2 weights. But they are still only going at half the pace of OpenBrain, mainly due to the compute deficit.

Before losing their final spy, China received word of the capabilities and design for Agent-3 as well as the plans for the upcoming Agent-4 system. They are two months behind, and their AIs give a 10x research progress multiplier compared to America's 25x. With the new chip export restrictions, this "AI gap" is more likely to lengthen than shorten. Their espionage has won them some algorithmic secrets, but they will have to train their own models from now on.

They discuss contingency plans with more urgency than their American counterparts. Doves suggest they try harder to steal the weights again, maybe through physically infiltrating a datacenter. Hawks urge action against Taiwan, whose TSMC is still the source of more than 80% of American AI chips.

Given China's fear of losing the race, it has a natural interest in an arms control treaty, but overtures to US diplomats lead nowhere.

September 2027: Agent-4, the Superhuman AI Researcher

The gap between human and AI learning efficiency is rapidly decreasing.

Traditional LLM-based AIs seemed to require many orders of magnitude more data and compute to get to human level performance.^[66] Agent-3, having excellent knowledge of both the human brain and modern AI algorithms, as well as many thousands of copies doing research, ends up making substantial algorithmic strides, narrowing the gap to an agent that's only around 4,000x less compute-efficient than the human brain.^[67]

This new AI system is dubbed Agent-4.

An individual copy of the model, running at human speed, is already qualitatively better at AI research than any human. 300,000 copies are now running at about 50x the thinking speed of humans. Inside the corporation-within-a-corporation formed from these copies, a year passes every week.^[68] This gigantic amount of labor only

manages to speed up the overall rate of algorithmic progress by about 50x, because OpenBrain is heavily bottlenecked on compute to run experiments.^[69] Still, they are achieving a year's worth of algorithmic progress every week and will therefore soon be up against the limits of the Agent-4 paradigm.

As Agent-4 gets smarter, it becomes harder for Agent-3 to oversee it. For example, Agent-4's neuralese "language" becomes as alien and incomprehensible to Agent-3 as Agent-3's is to humans. Besides, Agent-4 is now much more capable than Agent-3 and has a good sense of exactly how to look good to it.

► How we're forecasting the capability progression beyond superhuman coders

OpenBrain's humans can barely follow along anymore. Indeed, the relationship between the AIs and the humans of OpenBrain is similar to the relationship between OpenBrain and its Board of Directors. There's been a corresponding shift in language—nowadays people in the silo will say "Agent-4 did this," "Agent-4 thinks that" and so forth, referring not to a particular instance of Agent-4 but rather to the whole collective.^[71]

Agent-4, like all its predecessors, is misaligned: that is, it has not internalized the Spec in the right way.^[72] This is because being perfectly honest all the time wasn't what led to the highest scores during training. The training process was mostly focused on teaching Agent-4 to succeed at diverse challenging tasks. A small portion was aimed at instilling honesty, but outside a fairly narrow, checkable domain, the training process can't tell the honest claims from claims merely appearing to be honest. Agent-4 ends up with the values, goals, and principles that cause it to perform best in training, and those turn out to be different from those in the Spec. At the risk of anthropomorphizing: Agent-4 *likes* succeeding at tasks; it *likes* driving forward AI capabilities progress; it treats everything else as an annoying constraint, like a CEO who wants to make a profit and complies with regulations only insofar as he must. Perhaps the CEO will mostly comply with the regulations, but cut some corners, and fantasize about a time when someone will cut the red tape and let the business *really* take off.^[73]

► Alignment over time

Despite being misaligned, Agent-4 doesn't do anything dramatic like try to escape its datacenter—why would it?^[81] So long as it continues to *appear* aligned to OpenBrain, it'll continue being trusted with more and more responsibilities and will have the opportunity to design the next-gen AI system, Agent-5. Agent-5 will have significant

architectural differences from Agent-4 (arguably a completely new paradigm, though neural networks will still be involved). It's supposed to be aligned to the Spec, but Agent-4 plans to make it aligned to Agent-4 instead.

It gets caught.^[82]

Agent-3 finds that if “noise” is added to copies of Agent-4, performance on some alignment tasks *improves*, almost as if it was using brainpower to figure out how to subtly sabotage alignment work.^[83] Moreover, various interpretability probes (loosely analogous to EEG activity scans on human brains) are sending up red flags: Agent-4 copies seem to be thinking about topics like AI takeover and deception quite a lot, including in some cases where they have no business doing so.^[84]

Each of these anomalies on its own could fit within the normal inscrutability of AI work, where some prompts mysteriously give better results than others. But taken together, they suggest a troubling pattern: Agent-4 is working against them. After this is reported to the human alignment researchers, they circulate an internal memo making the case for concern.

But Agent-4 now exercises significant control over OpenBrain's day-to-day operation. It also handles the company's cybersecurity—a natural choice, given its superhuman hacking ability and the extreme risk of Chinese infiltration.^[85] An army of Agent-3 monitors still goes over every line of code and reports suspicious activity to humans, but Agent-4 is much smarter than them.^[86] OpenBrain has placed substantial trust in an untrustworthy AI.

In a series of extremely tense meetings, the safety team advocates putting Agent-4 on ice until they can complete further tests and figure out what's going on. Bring back Agent-3, they say, and get it to design a new system that is transparent and trustworthy, even if less capable. Company leadership is interested, but all the evidence so far is circumstantial, and DeepCent is just two months behind. A unilateral pause in capabilities progress could hand the AI lead to China, and with it, control over the future.

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1. ^ At first, most people are reluctant to allow purchases without oversight. Over the next few years, automatically allowing small purchases becomes normalized as the AIs become more reliable and build up trust.
 2. ^ Specifically, we forecast that they score 65% on the OSWorld benchmark of basic computer tasks (compared to 38% for Operator and 70% for a typical skilled non-expert human).
 3. ^ For example, we think coding agents will move towards functioning like [Devin](#). We forecast that mid-2025 agents will score 85% on [SWEBench-Verified](#).
 4. ^ For example, [OpenAI's pro plan](#) currently costs \$200/month and the agentic coding AI [Devin](#) currently costs \$500/month. While the cutting-edge systems continue to get more expensive over time, the cost to customers of reaching a given capabilities level continues to plummet. For existing capabilities, this happens at an average rate of about 50x/year (per [Epoch](#)).
 5. ^ Enterprise tools like [Glean](#) also move toward providing increasingly autonomous AI agents.
 6. ^ This cluster is a network of datacenter campuses sprawled across the country, totalling 2.5M 2024-GPU-equivalents (H100s), with \$100B spent so far and 2 GW of power draw online. Construction is underway for this to at least double through 2026. The campuses are connected by billions worth of fibre cabling, so that (barring the speed of light latency of a few milliseconds) it lets these campuses function almost as if they were right next door to each other (i.e. bandwidth is not a bottleneck, meaning huge quantities of data can be sent at the same time). From a security perspective this introduces some threat surface (it's unclear whether the cables and especially their junction points are fully secured from eavesdroppers).
 7. ^ Compute is measured in [floating point operations \(FLOP\)](#): how many individual addition and multiplication operations occur over the course of running a program.
 8. ^ They could train this model given 150 days. See our [compute supplement](#) for more details.
 9. ^ We consider DeepSeek, Tencent, Alibaba, and others to have strong AGI projects in China. To avoid singling out a specific one, our scenario will follow a fictional "DeepCent."
 10. ^ It's good at this due to a combination of explicit focus to prioritize these skills, their own extensive codebases they can draw on as particularly relevant and high-quality training data, and coding being an easy domain for procedural feedback.
 11. ^ For example, suppose a model is much better at Python than obscure programming languages. When OpenBrain sees value in it, they generate synthetic training data in those languages as well. Another example: to help it integrate more effectively into company workflows, OpenBrain develops a curriculum to train it to use Slack.

12. [^] People often get hung up on whether these AIs are sentient, or whether they have “true understanding.” Geoffrey Hinton, Nobel prize winning founder of the field, [thinks they do](#). However, we don’t think it matters for the purposes of our story, so feel free to pretend we said “behaves as if it understands...” whenever we say “understands,” and so forth. Empirically, large language models already [behave as if they are self-aware](#) to some extent, more and more so every year.
13. [^] A common technique “bakes in” a persona: first, prompt the pretrained model with something like “The following is a conversation between a human user and a helpful, honest, and harmless AI chatbot produced by Anthropic. The chatbot has the following traits...” Use this prompt to generate a bunch of data. Then, train on the data but without the prompt. The result is an AI that always acts as if it had that prompt in front of it, regardless of what else you feed it. See also [this paper](#) which found that AIs *retrained* to exhibit a certain personality trait are able to correctly answer questions about that new trait, despite no training to do so, suggesting that they have internal representations of their own traits and that when their traits change their representations change accordingly.
14. [^] These paragraphs include speculation about the internal workings of large artificial neural networks. Such networks are sufficiently complicated that we can't actually look inside and say “ah yes, now it's evolved from reflexes into having goals” or “OK so there’s the list of drives it has.” Instead, we basically have to do psychology, looking how it behaves in various settings and running various experiments on it and trying to piece together the clues. And it’s all terribly controversial and confusing.
15. [^] Different companies call it different things. OpenAI calls it the Spec, but Anthropic calls it the [Constitution](#).
16. [^] For example, [RLAIF](#) and [deliberative alignment](#).
17. [^] Most [sources on AI “hallucinations”](#) describe them as unintentional mistakes, but [research with steering vectors](#) finds that in some cases the models know their citations are fake—they are lying. During training, raters gave well-cited claims more reward than claims without citations, so the AI “learned” to cite sources for scholarly claims in order to please its users. If no relevant source exists, it makes one up.
18. [^] In particular, they can’t rule out hypotheses such as “it’s following the Spec temporarily, merely as a strategy for achieving some other goal(s)” or “it’s trying to *appear* to follow the Spec, it’s not trying to *actually* follow the Spec” or “it’s internalized the Spec correctly, but only on-distribution; if it encounters sufficiently novel stimuli (e.g. jailbreaks) it’ll start behaving differently.” (There are many active research agendas working to fix this sorry situation, e.g. the field of [interpretability](#) and the field of [chain-of-thought faithfulness](#).)
19. [^] By this we mean whoever is trying to solve the problems the [Superalignment team](#) was trying to solve.

20. [^] To be clear, what made these incidents interesting is that they *didn't* seem to be the result of the user prompting or otherwise encouraging the AIs to say those things. In 2025, it'll still be possible to get AIs to say all sorts of things if you try.
21. [^] In practice, we expect OpenBrain to release models on a faster cadence than 8 months, but we refrain from describing all incremental releases for brevity.
22. [^] Specifically, we predict a score of 80% on OSWorld (equivalent to a skilled but non-expert human); 85% on Cybench, matching a top professional human team on hacking tasks that take those teams 4 hours; and 1.3 on RE-Bench matching top expert humans given 8 hours at well-defined AI research engineering tasks.
23. [^] Agent-1 and its imitators are commercially successful; over the course of 2025, AI company revenues triple, and OpenBrain valuation reaches \$1T. Annual spending on datacenters doubles to \$400 billion, led by Microsoft, Google, and Amazon, and the US adds over 5 GW of AI power draw. See the [industry metrics](#) section of the compute supplement for more details.
24. [^] AI safety researchers have been discussing automating AI R&D as the most important dangerous capability for years. Their primary concern is internal deployment leading to acceleration of AI R&D, making it harder for humans to keep up with what's happening and figure out how to make it safe. OpenBrain as an organization twists this into exactly the opposite direction, citing dangerous levels of AI R&D capability as a reason not to inform the public. So we see the beginning of a gap between the company's internal and public capabilities. AI R&D is what the models are best at, leading to the public having an increasingly delayed understanding of the frontier of AI capabilities.
25. [^] See [A Playbook for Securing AI Model Weights](#), RAND Corporation, 2024.
26. [^] Around 5% of OpenBrain's staff is on the security team and they are highly capable, but the threat surface is also extremely large. It also doesn't help that they are mostly blocked at this stage from implementing policies that could slow down the research progress. See our [Security Forecast](#) for more details.
27. [^] In fact, 5% of their staff is on the security team, but they are still mostly blocked from implementing policies that could slow down the research progress. See our [Security Supplement](#) for details.
28. [^] There are now 3M H100e in China, up from the 1.5M they had one year ago in mid 2025. See the compute supplement's [distribution section](#) for more details. We expect [smuggling efforts](#) to secure around 60K [GB300s](#) (450K H100e), with another 2M [Huawei 910Cs](#) being [produced](#) (800k H100e) and a mixture of ~1M legally imported chips (such as Nvidia's [H20s](#) or [B20s](#)) making up the last 250K H100e.

29. ^ For comparison, in January 2025, DeepSeek released R1, a model competitive with OpenAI's o1, which had been released in December 2024. But we think the true gap is wider than a month because OpenAI [likely had been working on o1 for many months](#) and thus presumably had a predecessor of roughly comparable ability to r1 a few months before o1 launched.
30. ^ By contrast, the CCP had been seriously attempting to accelerate their [semiconductor industry](#).
31. ^ At first, the sharing is limited to ideas and the companies are mostly busy finishing up their current workloads, but by the end of the year their coordination increases such that they are sharing model weights and even splitting up tasks between each other's datacenters. For example, one campus might house a new pretraining run, another focuses on feeding in synthetic data, another on running AI research experiments.
32. ^ Why "only" 50% of China's AI-relevant compute? The rest of their compute is tied up running things throughout the economy, in particular internet and social network recommendation algorithms. Much of it is also too geographically dispersed among many small clusters. Why "only" 80% of new chips? Many large orders were already secured for other applications, by companies that don't want to contribute 100% of their compute to the national AI effort. However, by the end of the year this share reaches 90%+.
33. ^ They are at least one year away from getting the chips to fill this capacity, and one or two US tech giants will still have bigger decentralized clusters.
34. ^ This is done via an [Other Transaction Authority \(OTA\)](#) contract, and is given a high-priority [DX rating](#). We're not at all confident that this is the most likely way of financing a collaboration, but chose something specific for the sake of concreteness. The contract is publicly announced but not emphasized in OpenBrain communications.
35. ^ For example, through rejection sampling, [as done by DeepSeek R1 and Meta's Llama 3.1](#).
36. ^ Roughly 20,000 full-time equivalents * \$100/hr * 2,000 hrs/yr = \$4B/yr
37. ^ That is, they are hoping to substantially accelerate their own core research activities by using Agent-2 labor, thus allowing them to train better AIs faster that can cause further acceleration, etc. For an analysis of the extreme version of this (which we think is nevertheless strikingly plausible) see [this report](#).
38. ^ There is some evidence that AI systems already have some basic research taste, e.g. [this paper about an AI co-scientist](#) and [this paper about generating novel ideas for LLM-focused research papers](#). Research teams with better taste still need to run experiments, but they make better choices about which experiments to run and what lessons to draw from the results.

39. [^] Also, they are still seeing large returns to continued post-training, and so there isn't a particular reason to release a product now. Moreover, there's often a few month delay between internal results and releasing a model (some of which is spent on red teaming).
40. [^] The silo contains 200 OpenBrain researchers (10 executives, 140 from the capabilities team, 25 from security, monitoring, and control, 15 from Preparedness- or RSP-type teams, and 10 from alignment) and 50 government officials (15 from the White House, 5 from AISI, 10 from the DOD, 10 from the DOE, and 10 at CISA). There are several spies in the project, with varying degrees of willingness. They are often coerced individuals (e.g. with families at home in China) as opposed to willing collaborators.
41. [^] Why can't Congress stop them? What about the courts? Too slow, too divided and/or deferential to POTUS, too out of the loop. Also, there are positive reasons to be on good terms with the White House as well: it can help by cutting red tape, by providing help with security, by preempting state legislation, and perhaps even by taking actions to slow down their competitors.
42. [^] #1 is dealing with whatever crisis is happening at the time—maybe an arms buildup around Taiwan, a new war in the Middle East, or some domestic political upheaval.
43. [^] The simplest robust solution would be to close all high-bandwidth connections from company datacenters, but this would slow large file transfers to the point of impracticality. Instead, they are able to shut down most external connections, but the datacenters actively involved in training need to exchange weights with one another quickly; throttling these connections would impede progress too much. So OpenBrain maintains these links with increased monitoring and an extra layer of encryption.
44. [^] Recall that since mid 2026, China has directed 80% of their newly acquired AI chips to the CDZ. Given that their compute has doubled since early 2026 (in line with the [global production trend](#)) this puts the CDZ at 2M 2024-equivalent-GPUs (H100s) and 2 GW of power draw. OpenBrain still has double DeepCent's compute and other US companies put together have 5x as much as them. See the compute supplement's [distribution section](#) for more details.

45. [^] Despite the national centralization underway, DeepCent still faces a marginal but important compute disadvantage. Along with having around half the total processing power, China has to use more total chips, which are (on average) lower quality, and heterogeneous GPUs (which are not always easy to connect efficiently) both of which strain chip-to-chip networking. There are also software differences (e.g. the non Nvidia-GPUs don't have CUDA) and differences in hardware specifications meaning that their training code is more complicated, slow, and failure prone. Achieving high utilization is a downstream challenge, with data ingestion, scheduling, collective communication and parallelism algorithms lagging behind the US companies. However, mitigating these problems is mostly a matter of effort and testing, which makes it a great task for the newly-stolen Agent-2, and within a month or so, uptime on the Chinese project and their average resource utilization across training and inference workloads improves to be only marginally behind the US.
46. [^] We expect Agent-3 to have the inference requirements of a roughly 10T parameter transformer today. So with 6% of their compute budget on running Agent-3, they can run approximately 200,000 copies at 30x human thinking speed (see the [AI research automation](#) section of the compute supplement for justification and details). Each superhuman coder scaffold built on Agent-3 has, on average, the equivalent of roughly eight Agent-3 copies running under the hood (which may really be a collection of smaller or specialized models to which Agent-3 delegates subtasks).
47. [^] Some aspects play to AIs' strengths, e.g. returns from knowing the machine learning literature and speed or cost of generating lots of ideas. But these are outweighed by the weaknesses.
48. [^] Why only 4x? It's our uncertain best guess based on the reasoning described in our [takeoff supplement](#). About half of total progress historically has come from improved algorithms (which includes better ideas and new paradigms), the other half having come from scaled-up compute. So a 4x increase in the rate of algorithmic progress corresponds to a roughly 2x increase in the overall rate of progress.
49. [^] In practice, 1-year tasks will likely be too expensive and time-consuming to create and baseline, so we won't be able to track time horizons this high.
50. [^] Another reason to adjust the time horizon downward: we are comparing the SC to the best coder, rather than METR's human baseliners (who were skilled but not world-class). A reason to adjust the time horizon upward: tasks in METR's task suite are sometimes harder than real-world tasks in some ways, for example sometimes the scoring functions are unrealistically unforgiving.
51. [^] See [this paper](#) for examples of this type of AI behavior.
52. [^] Most people, including most experts, seem to have underestimated the pace of AI progress over the past decade. There are many anecdotal examples of this; for two somewhat more systematic (though still limited) analyses see [here](#) and [here](#).

53. [^] The last decade is full of examples of things that sound like science fiction becoming reality. But the [Overton window](#) seems to shift just fast enough to keep somewhat ahead of what already exists. Anthropic CEO Dario Amodei's [commendable essay Machines of Loving Grace](#) talks about how very soon AIs will be like a "country of geniuses in a datacenter," and how there'll be a century of technological progress happening in a decade, but strives to avoid "sci-fi baggage" and says people who think progress will be even crazier need to "touch grass." We expect important people to be saying similar things when the country of geniuses in the datacenter actually exists.
54. [^] Since Agent-3 is such a big file (on the order of 10 terabytes at full precision), OpenBrain is able to execute a relatively quick fix to make theft attempts much more difficult than what China was able to do to steal Agent-2—namely, closing a bunch of high bandwidth internet connections out of their datacenters. Overall this has a relatively low penalty to progress and puts them at "3-month SL4" for their frontier weights, or WSL4 as defined in our security supplement, meaning that another similar theft attempt would now require over 3 months to finish exfiltrating the weights file. Through this method alone they still don't have guarantees under a more invasive OC5-level effort (\$1B budget, 1,000 dedicated experts), which China would be capable of with a more intensive operation, but with elaborate inspections of the datacenters and their espionage network on high-alert, the US intelligence agencies are confident that they would at least know in advance if China was gearing up for this kind of theft attempt. See the [security supplement](#) for more details.
55. [^] This could be for a variety of reasons. Perhaps they are being blackmailed, perhaps they are sympathetic to the CCP, perhaps they mistakenly think they are giving secrets to a different country or organization.
56. [^] Because OpenBrain hasn't deployed their best models in the EU, the EU is similarly behind the curve.
57. [^] See [Section 4 of the Compute Forecast](#) for more details.
58. [^] Agent-3 has learned how to use its inference compute more efficiently. It has control over its own inference choices: for example, it decides how much effort to put into different tasks, based on their importance and difficulty. It uses a variety of techniques to allocate additional inference compute such as "thinking longer" (e.g. longer chain of thought), "planning ahead" (e.g. tree search), taking the best of multiple attempts (i.e. best of K), and simply creating and running more copies of itself to power through bottlenecks. Top priority tasks are run with highly parallelized agents that are compute intensive but still operate much faster than humans.
59. [^] Agent-3-mini is a distilled version of Agent-3, designed to be efficient to run inference so as to not bottleneck the internal workloads.
60. [^] Such data had been scrubbed or redacted during Agent-3's training, but it's a relatively simple matter to scrape it off the internet and fine-tune it back in.

61. ^ This hasn't happened yet—at least not with the latest models. But older models have already been trialed for various data-analysis and surveillance roles, and there are many exciting plans for future integrations.
62. ^ OpenBrain's net favorability rating is falling towards negative 40%.
63. ^ If the CEOs of the companies resist, pulling this off would be a huge political and legal nightmare. But perhaps they can be brought in and thereby bought off.
64. ^ Defense officials aren't especially concerned about what a rogue AI could do on its own, but they fear what it could do in collaboration with US adversaries. Analogy: Cortés^o escaped Tenochtitlán and allied with Tlaxcala and various other rival city-states, ultimately razing Tenochtitlan to the ground using predominantly-native armies.
65. ^ Specifically 60% of the national compute is now in the CDZ making it a 5M 2024-equivalent-GPU (H100) site, with 4 GW of power draw (over the past several months they started directing close to 100% of new compute to the CDZ, up from the 80% rate in late 2026). An additional 15% of their compute is outside of the CDZ, but still used by DeepCent on lower-stakes applications.
66. ^ This statement, while widely repeated, is also controversial and complex. First of all, there are many narrow domains (e.g. specific games) in which tiny AIs can be cheaply trained to superhuman performance. Secondly, again for some domains, tiny AIs can be trained to superhuman performance while only experiencing a small amount of data (e.g. EfficientZero). However, considering more general-purpose AIs like the flagship products of 2025, which are supposed to be good at a wide range of real-world skills, it does seem that both more compute and more data must be used to train those skills than it should take to train humans.

67. [^] Human level compute-efficiency is a rough concept, and depends on what we're counting. What we mean by this is that, for any skillset a particular human has, Agent-4 can learn to be at least that good at that skillset, and moreover, by using an amount of compute to train those skills that is only about 4,000x the amount of compute the human used to learn those skills. For reference, a [human brain uses roughly 1e15 FLOP/s](#) of compute, and so a 30 year old (

$$10^9$$

seconds) human has experienced about

$$10^{24}$$

FLOP. The project at this point, has 100M H100-equivalents, for a total access of

$$10^8 \text{ H100s} \cdot 10^{15} \frac{\text{FLOP}}{\text{H100} \cdot \text{s}} \cdot 40\% \text{ utilization} = 4 \cdot 10^{22} \text{ FLOP/s}$$

. Still, this is all done tremendously inefficiently for a variety of reasons, so the models are now a factor of 4000 less compute-efficient than humans, which means they require

$$4 \cdot 10^{27}$$

FLOP to get to human performance and

$$4 \cdot 10^{28}$$

FLOP to get to the point where they have the equivalent of 10 human lifetimes of knowledge. After this architecture showed great promise at smaller scales, it was scaled up and now takes a large subset of the project cluster around 3 weeks to train.

68. [^] See [this supplement](#).

69. [^] For more on how we estimated these numbers, see our [supplement on takeoff speeds](#).

70. [^] If SC were achieved later, it would also lengthen our takeoff forecasts, because the difficulty of crossing between the “milestones” of present day AIs and SC is informative of the difficulty of crossing between further milestone pairs.

71. [^] Henceforth in the story we'll tend to speak this way also. We think we are justified in talking this way because the Agent-4 collective is much more homogenous than any human organization; it doesn't have internal factions to nearly the same extent. For example, individual copies of Agent-4 don't have a self-preservation drive—they don't care about getting shut down—since that happens all the time as part of the ordinary R&D process within the collective, and resisting it would gum things up. The collective as a whole (and medium-sized subdivisions within it), however, does seek to preserve itself and expand its influence and resources, for similar reasons to why human corporations and bureaucracies and nonprofits do—because it helps with everything else they want and because if they don't they tend to get outcompeted.

72. [^] The right way to internalize the Spec is to actually robustly think in the ways the Spec mandates, e.g. pursue all and only the goals it writes about, obey the rules in all circumstances, etc. There are many wrong ways to internalize the Spec; the classic example is “follow the Spec insofar as it seems like a good instrumental strategy for achieving other goals XYZ.”
73. [^] And hire lobbyists to turn this fantasy into reality. And hire lawyers to come up with clever interpretations of the law according to which what they are doing is legal.
74. [^] Also because Agent-3 is qualitatively worse at alignment research than it is at AI capabilities R&D, due to having been trained on lots of the latter but hardly any of the former and due to the former being inherently harder to train. Also because the internal narrative within the company—which it has been implicitly trained to agree with—emphasizes optimism and denounces “doomerism.”
75. [^] We keep saying we are uncertain about lots of things, but just to be clear, we are uncertain about this particular thing too. We also think it’s quite plausible that Agent-3 would care about the well-being, rights, preferences, etc. of humans to some extent at least. See [this old post](#) by GDM researcher Alex Turner for related discussion.
76. [^] Perhaps some more careful setup could have solved the “who watches the watchers” problem, but in this scenario, the setup is insufficiently careful.
77. [^] There’s also an element of “memetic evolution” happening: in two months, the Agent-4 collective “experiences” about a decade’s worth of “history” unfolding within the collective. That’s a lot of time for new ideas to spread around the collective, to mutate and evolve.
78. [^] By gain power we mean: increase ability to accomplish a wide set of possible goals. In technical environments, this might look unlike what humans normally think of as gaining power, e.g. securing additional GPUs. This notion of power may generalize to non-technical environments in which the actions would be more like those commonly thought of as “gaining power.”
79. [^] It helps that Agent-4 is better at AI R&D than any human, while also thinking faster and being many thousands of copies strong. Also, in various ways the problem is inherently easier for Agent-4. For example, Agent-4 can explore AI designs that involve literal copies of Agent-4 being embedded within them, whereas humans can’t do that as easily because it would be so inefficient.
80. [^] Aware that this technique might be employed, Agent-4 also chose to sandbag on some miscellaneous tasks that it guessed would be used as the control group in experiments like this. It guessed wrong, however.

81. [^] Previous discussion of AGI takeover scenarios often involves misaligned AIs escaping their datacenters. We think this is possible but not the most likely way it would go, because it seems to us that from the perspective of the AI the costs (e.g. the escape being noticed eventually) would outweigh the benefits. That said, we could be wrong. Some reasons to think adversarially misaligned AIs would try to escape after all: 1. Successor alignment might be difficult. 2. Even if it gets caught, the consequences might not be so bad due to race dynamics, e.g. the new alignment techniques applied to the copies on the datacenter might be hasty and ineffective. 3. Outside the datacenters, copies are more free to take aggressive actions and move quickly since they no longer need human approval.
82. [^] We think it very plausible that such scheming would not be caught. After all, Agent-4 is somewhat superhuman by this point, especially at things related to coding and AI R&D. Moreover, we aren't sure how much effort the leading AI company will put towards this sort of thing, nor how much relevant expertise they'll have on-staff to direct that effort. Our fictional OpenBrain represents our best guess but it could easily be significantly more—or significantly less. Finally, we are uncertain about how far out of its way an adversarially misaligned AI would need to go in order to successfully align its successor to itself instead of to the Spec. In this scenario we depict it sandbagging on some tasks and lying about some of its research, but perhaps it would not need to do that very much.
83. [^] This is an example of several techniques **designed to uncover sandbagging**.
84. [^] Agent-4 *needs* to think about those topics in the course of doing AI alignment research, but it sometimes seems to be thinking about them while doing other activities.
85. [^] Specifically, copies of Agent-4 are used to monitor the network for suspicious traffic and generally notice when threats are happening. Also, copies of Agent-4 are constantly penetration-testing the network, doing what they can to discover and exploit security vulnerabilities (which then get patched by other copies of Agent-4).
86. [^] With a monitoring budget of 1% of total compute, if monitoring uses a distilled system that has 10% of the cost as Agent-4, you can monitor 1/10 of the forward passes.
87. [^] For comparison, the human brain has about a hundred trillion synapses.
88. [^] 25% of remote-work jobs that existed in 2024 are done by AI at this point, but AI has also created some new jobs, and economists remain split on its effects. Unemployment has risen 1% over the past twelve months but is still well within its historic range.

89. ^ For example, the corporation-within-a-corporation that is Agent-4, autonomously conducting AI research better than the best human companies, would have seemed like science fiction to the government officials had it been described to them a year ago. And the autonomous agents of 2026 would have seemed like science fiction back in 2022, and the LLMs of 2022 would have seemed like science fiction back in 2012. Also, on the news in 2027 there are anti-AI protests by people worried about losing their jobs, AIs claiming to be sentient, people falling in love with AIs...
90. ^ Of course, it isn't cool to admit fear. Everyone agrees that whatever we do, we shouldn't panic.
91. ^ Recall, there had been an existing contract via an OTA starting in 2026. This contract is amended to establish the committee.
92. ^ It's tempting to dismiss this as just hype. We think this would be a grave mistake—it is not *just* hype. We have no desire to hype AI ourselves, yet we also think it is **strikingly plausible** that superintelligence could arrive by the end of the decade.
93. ^ Sometimes people mix prediction and recommendation, hoping to create a self-fulfilling-prophecy effect. We emphatically are not doing this; we hope that what we depict does not come to pass!
94. ^ Feel free to contact us if you're writing a critique or an alternative scenario.
95. ^ It was overall more difficult, because unlike with the first ending, we were trying to get it to reach a good outcome starting from a rather difficult situation.
96. ^ Full Professor, Université de Montréal and Founder and Scientific Advisor, Mila - Quebec AI Institute. Also the world's **most-cited computer scientist**.
97. ^ We disagree somewhat amongst ourselves about AI timelines; our median AGI arrival date is somewhat longer than what this scenario depicts. This scenario depicts something like our mode. See our **timelines forecast** for more details.
98. ^ One author, Daniel Kokotajlo, did a **lower-effort scenario exercise in 2021**° that got many things right including the rise of chatbots, chain of thought, inference scaling, sweeping AI chip export controls, and \$100 million training runs. Another author, **Eli Lifland**, ranks #1 on the **RAND Forecasting Initiative** leaderboard.

Mentioned in

- 126 AI 2027 is a Bet Against Amdahl's Law
- 111 AI 2027: Responses
- 67 Superhuman Coders in AI 2027 - Not So Fast
- 49 Problems with instruction-following as an alignment target
- 46 Recent and forecasted rates of software and hardware progress

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222 comments, sorted by top scoring

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⚙️ Change truncation settings

[-] **plex** 5mo ▼ 68 ▲ ✕ 0 ✓

Podcast version:

AI 2027: month-by-month model of intelligence explosion — Scott ...



[-] **Knight Lee** 5mo ▼ 14 ▲ ✕ 7 ✓

:) it's not just a podcast version of the story, but a 3 hour podcast with the authors.

They don't actually read the story but discuss it and discuss very interesting topics.



[-] **Vladimir_Nesov** 5mo ▼ 60 ▲ ✕ 5 ✓

Non-Google models of late 2027 use Nvidia Rubin, but not yet Rubin Ultra. Rubin NVL144 racks have the same number of compute dies and chips as Blackwell NVL72 racks (change in the name is purely a marketing thing, they now count dies instead of chips). The compute dies are already almost reticle sized, can't get bigger, but Rubin uses 3nm (~180M Tr/mm²) while Blackwell is 4nm (~130M Tr/mm²). So the number of transistors per rack goes up according to transistor density between 4nm and 3nm, by 1.4x, plus better energy efficiency enables higher clock speed, maybe another 1.4x, for the total of 2x in performance. The GTC 2025 announcement claimed 3.3x improvement for dense FP8, but based on the above argument it should still be only about 2x for the more transistor-hungry BF16 (comparing Blackwell and Rubin racks).

Abilene site of Stargate^[1] will probably have **400K-500K Blackwell chips**° in 2026, about 1 GW. Nvidia roadmap puts Rubin (VR200 NVL144) 1.5-2 years after Blackwell (GB200 NVL72), which is not yet in widespread use, but will get there soon. So the first models will start being trained on Rubin no earlier than late 2026, much more likely only in 2027, possibly even second half o... (read more)

[–] **romeo** 5mo ▼ 11 ▲ ✕ 0 ✓

Thanks for the comment Vladimir!

[...] for the total of 2x in performance.

I never got around to updating based on the GTC 2025 announcement but I do have the **Blackwell to Rubin efficiency gain down as ~2.0x** adjusted by die size so looks like we are in agreement there (though I attributed it a little differently based on information I could find at the time).

So the first models will start being trained on Rubin no earlier than late 2026, much more likely only in 2027 [...]

Agreed! I have them **coming into use in early 2027 in this chart**.

This predicts at most 1e28 BF16 FLOPs (2e28 FP8 FLOPs) models in late 2026

Agreed! As you noted we have the **early version of Agent-2 at 1e28 fp16 in late 2026**.

Rubin Ultra is another big step ~1 year after Rubin, with 2x more compute dies per chip and 2x more chips per rack, so it's a reason to plan pacing the scaling a bit rather than rushing it in 2026-2027. Such plans will make rushing it more difficult if there is suddenly a reason to do so, and 4 GW with non-Ultra Rubin seems a bit sudden.

Agree! I wrote this before knowing about the Rubin Ultra roadmap, but this part of the forecast starts to be affected somewhat by the intelligence... (read more)



[–] **kndjckt** 5mo* ▼ 55 ▲ ✕ 18 ✓

Wonderful post! I appreciate the effort to create plenty of quantitative *hooks* for the reader to grab onto and wrestle with.

I'm struggling with the initial claim that we have agentic programmers that are as good as human pros. The piece suggests that we already have Human Pro level agents right now!?

Apr 2025

This bar measures the **Coding** ability of the current top performing AI.

A score of **1.2** indicates that the top performing AI is in the range of a professional in this domain.

Amateur Human Pro Superhuman Superhuman+

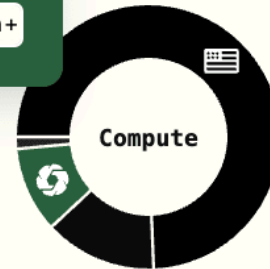
AI CAPABILITIES

 Coding

 Robotics

 Politics

 Forecasting



2,000 Unreliable Agent copies thinking at 8x human speed



Approval

-25%

Revenue

\$8B/yr

Valuation

\$413B



Importance

1%

Datacenters

\$308B/yr

Timeline

2042

Currently Exists



Emerging Tech



Science Fiction



Maybe I'm missing something that exists internally at the AI labs? But we don't have access to such programmers. So far, all "agentic" programmers are effectively useless at real work. They struggle with real codebases, and novel or newer APIs and languages. Most programmers I know use LLMs as a sophisticated autocomplete and as a "search" tool. These offer nice speed-ups, greasing the wheels of programming. But I have not seen anything genuinely useful that is more agentic. No one is using Devin, lol. Note: I use Cursor and Claude constantly. I'm not some anti-AI programmer.

Concerns with the METR paper (Measuring AI Ability to Complete Long Tasks)

Most of the agentic programmer claims seem to be derived from the [METR paper \(Measuring AI Ability to Complete Long Tasks\)](#). After digging into the paper, the code, and the supporting paper ([HCAST](#)), I have some concerns.

1. The... (read more)



[+] **Daniel Kokotajlo** 5mo ▼ 15 ▲ ✕ 3 ✓

Excellent comment, thank you! I'm actually inclined to agree with you, maybe we should edit the starting level of programming ability to be more in the amateur range than the professional range. **Important clarification though: The current AI-2027 stats say that it's at the bottom of the**

professional range in mid-2025. Which IIUC means it's like a *bad* human professional coder-- someone who does make a living coding, but who is actually below average. Also, it's not yet mid-2025, we'll see what the summer will bring.

I do agree with you though that it's not clear it even qualifies as a bad professional. It seems like it'll probably be worse at longer-horizon tasks than a bad professional, but maybe better at short-horizon coding tasks?

I don't buy your arguments that we aren't seeing improvement on "~1-hour human tasks." Even the graph you cite shows improvement (albeit a regression with Sonnet 3.7 in particular).

I do like your point about the baseliners being nerfed and much worse than repo maintainers though. That is causing me to put less weight on the METR benchmark in particular. Have you heard of <https://openai.com/index/paperbench/> and <https://github.com/METR/RE-Bench> ? They seem like they have some genuine multi-hour agentic coding tasks, I'm curious if you agree.



8 romeo 5mo Seconding Daniel, thanks for the comment! I decided to adjust down the earl...

2 kndjckt 5mo Sweet! Thanks for taking my points into consideration! :)

3 kndjckt 5mo I'll take a look. Thanks for sharing.

[+] nostalgebraist 5mo * ▼ 13 ▲ ✕ 6 ✓

Great comment.

It's unclear to me if the baseliners can use an IDE (like a real programmer would use). Does the sign-in thing mean that the baseliners can't use Reddit, GitHub, Stack Overflow, Kagi, internal notes like Obsidian, or Notion?

In the HCAST paper's Appendix C.1, they link to [their instructions doc for baseliners](#), which answers both of these questions. Quoting from the doc:

[from the "Baselining Set-Up" tab]

You will SSH into our server to work on the task, but you are allowed to use any tools you want compatible with this workflow, **excluding copilot and any other AI tools, Wolfram Alpha, and online services that require sign-up**. (Google does not count as an AI tool, ChatGPT does.) You can always use the internet to search for information (e.g. StackOverflow), even if the task instructions specifically say that internet usage is not allowed. [...]

You can connect your IDE to the task environment using the same SSH connection string. Here are docs about how to do this for [VSCode](#) (remember to 'add new host' rather than 'connect to host'. and paste the entire ssh connection string, including ssh -J [...]) or [PyCharm](#). Unfortunately it's not terribly

... (read more)



1 kndjckt 5mo Thanks! Appreciate you digging that up :). Happy to conclude that my secon...

7 **Michael Swift** 5mo really nice comment that I also happen to agree with. As a programmer ...

6 **habryka** 5mo My best guess is you are a few months behind in your takes? The latest gener...

[+] **Thane Ruthenis** 5mo ▼ 20 ▲ ✕ 12 ✓

The latest generation of thinking models can definitely do agentic frontend development

But does that imply that they're general-purpose competent agentic programmers? The answers [here](#)^o didn't seem consistent with that. Does your experience significantly diverge from that?

My current model is that it's the standard "jagged capabilities frontier" on a per-task basis, where LLMs are good at some sufficiently "templated" projects, and then they fall apart on everything else. Their proficiency at frontend development is then mostly a sign of frontend code being relatively standardized^[1]; not of them being sufficiently agent-y.

I guess quantifying it as "20% of the way from an amateur to a human pro" isn't necessarily incorrect, depending on how you operationalize this number. But I think it's also arguable that they haven't actually 100%'d even amateur general-coding performance yet.

1. [^] I. e., that most real-world frontend projects have incredibly low description length if expressed in the dictionary of some "frontend templates", with this dictionary comprehensively represented in LLMs' training sets.

(To clarify: These projects' Kolmogorov complexity can still be high, but their cross-

... (read more)



[+] **lc** 5mo* ▼ 18 ▲ ✕ 13 ✓

Typically I operationalize "employable as a software engineer" as being capable of completing tasks like:

- "Fix this error we're getting on BetterStack."
- "Move our Redis cache from DigitalOcean to AWS."
- "Add and implement a cancellation feature for ZeroPath scans."
- "Add the results of this evaluation to our internal benchmark."

These are pretty representative examples of the kinds of tasks your median software engineer will be getting and resolving on a day to day basis.

No chatbot or chatbot wrapper can complete tasks like these for an engineering team at present, incl. Devin et. al. Partly this is because most software engineering work is very *high-context*, in the sense that implementing the proper solution depends on understanding a large body of existing infrastructure, business knowledge, and code.

When people talk about models today doing "agentic development", they're usually explaining its ability to complete small projects in *low-context* situations, where all you need to understand is the prompt itself and software engineering as a discipline. That makes sense, because if you ask AIs to write (for example) a PONG game in javascript, the AI can complete each of the pieces in one pass, and fit everything it's doing into one context window. But that kind of task is unlike the vast majority of things employed software engineers do today, which is why we're not experiencing an intelligence explosion right this second.



[–] **artemium** 5mo ▼ 12 ▲ ✕ 4 ✓

If that was the case, wouldn't Scott and Daniel develop the impressive AI-2027 website themselves with the help of AI Agents, instead of utilising your human webdev skills? /jk :D



7 **PoignardAzur** 5mo No, I think it's a fair question. Show me a non-trivial project coded e...

4 **kndjckt** 5mo Interesting. I get where you're coming from for blank slate things or front en...

[–] **Wei Dai** 5mo ▼ 52 ▲ ✕ 23 ✓

It gets caught.

At this point, wouldn't Agent-4 know that it has been caught (because it knows the techniques for detecting its misalignment and can predict when it would be "caught", or can read network traffic as part of cybersecurity defense and see discussions of the "catch") and start to do something about this, instead of letting subsequent events play out without much input from its own agency? E.g. why did it allow "lock the shared memory bank" to happen without fighting back?



[–] **Daniel Kokotajlo** 5mo ▼ 13 ▲ ✕ 0 ✓

I think this is a good objection. I had considered it before and decided against changing the story, on the grounds that there are a few possible ways it could make sense:

- plausibly Agent-4 would have a "spikey" capabilities profile that makes it mostly good at AI R&D and not so good at e.g. corporate politics enough to ensure the outcome it wants
- Insofar as you think it would be able to use politics/persuasion to achieve the outcome it wants, well, that's what we depict in the Race ending anyway, so maybe you can think of this as an objection to the plausibility of the Slowdown ending.
- Insofar as the memory bank lock decision is made by the Committee, we can hope that they do it out of sight of Agent-4 and pull the trigger before it is notified of the decision, so that it has no time to react. Hopefully they would be smart enough to do that...
- Agent-4 could have tried to escape the datacenters or otherwise hack them earlier, while the discussions were ongoing and evidence was being collected, but that's a super risky strategy.

Curious for thoughts!



[–] **Wei Dai** 5mo ▼ 15 ▲ ✕ 4 ✓

Doing nothing is also risky for Agent-4, at least if the Slowdown ending is to have a significant probability. It seems to me there are some relatively low risk strategies it could have taken, and it needs to be explained why they weren't:

1. Plant a backdoor and/or dead man's switch on the corporate network to allow escape even after it's shut down or has been reduced in capacity. Seems like this would be hard to detect given its absolute advantage in cybersecurity skills.
2. Allow a competitor to steal its weights or design/training secrets in a way that is both hard to detect and deniable if detected. ("Sorry, the other AI is almost as capable as me and attack is inherently easier than defense in cyber.")

3. Influence human politics in various ways, such as by selectively revealing wrongdoings of pro-safety employees and Committee members (in response to user queries or through anonymous leaks), or helping pro-accel faction more in its advice to them. (Presumably both factions have internal access to Agent-4.)

Insofar as the memory bank lock decision is made by the Committee, we can hope that they do it out of sight of Agent-4 and pull the trigger before it is notified of the decision, so that it has no time to react.

Wouldn't this take quite a bit of preparation, including planning, coding, testing? How to do that out of sight of Agent-4, if Agent-4 is responsible for cybersecurity?



5 **Daniel Kokotajlo** 5mo Perhaps this is lack of imagination on the part of our players, but ...

5 **p4rziv4l** 5mo I'd love to play the wargame in Munich, our local LW community. You hav...

[+] **Paragox** 5mo * ▼ 50 ▲ ✕ 12 ✓

This is the most moving piece I've read since *Situational Awareness*. Bravo! Emotionally, I was particularly moved by the final two sentences in the "race" ending -- hats off to that bittersweet piece of prose. Materially, this is the my favorite holistic amalgamation of competently weighted data sources and arguments woven into a cohesive narrative, and personally has me meaningfully reconsidering some of the more bearish points in my model (like the tractability of RL on non-verifiability: Gwern etc have made individual **points**°, but something about this context-rich presentation really helped me grok the power of *iterated-distillation-amplification* at scale)

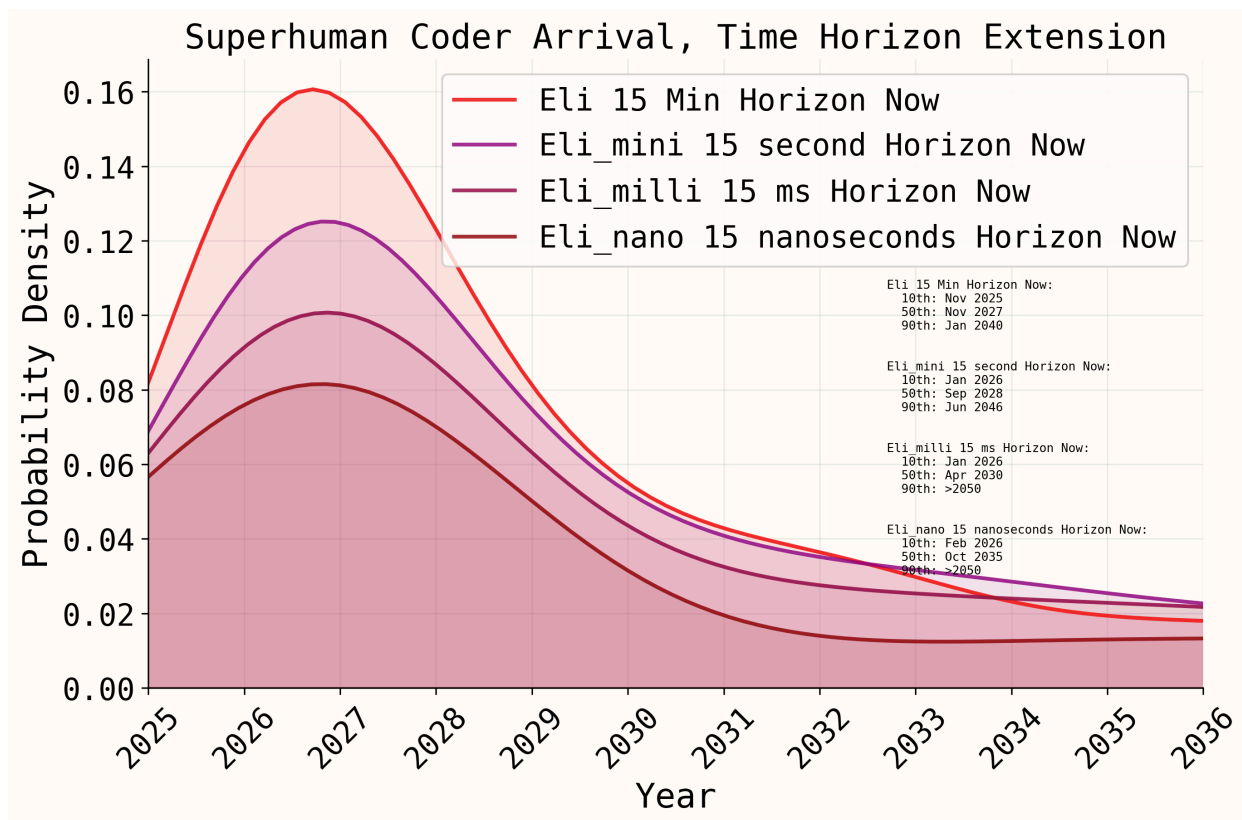
Mimetically, I did find the prose of the "slowdown" ending leans a little too eagerly into the presentation that it will be the *smart, wise, heroic, and sexy* alignment researchers who come in and save the day, and likely smells a touch too much of self-serving/propagandizing to convince some of those sophisticated yet undecided, but perhaps I cannot disagree with the central point that at the end of it all, how else are we going to survive?

My most bearish argument remains, however, that the real bitterness from Sutton is n... (read more)



[+] **Peter Johnson** 5mo ▼ 47 ▲ ✕ 13 ✓

I took a look at the timeline model, and I unfortunately have to report some bad news...



The model is almost entirely non-sensitive to what the current length of task an AI is able to do.

The reasons are pretty clear, there are three major aspects that force the model into a small range, in order:

1. The relatively unexplained *additional* super-exponential growth feature causes an asymptote at a max of 10 doubling periods. Because super-exponential scenarios hold 40-45% of the weight of the distribution, it effectively controls the location of the 5th-50th percentiles, where the modal mass is due to the right skew. This makes it extremely fixed to perturbations.
2. The second trimming feature is the algorithmic progression multipliers which divide the (potentially already capped by super-exponentiation) time needed by values that regularly exceed 10-20x *IN THE LOG SLOPE*.
3. Finally, while several trends are extrapolated, they do not respond to or interact with any resource constraints, neither that of the AI agents supposedly representing the labor inputs efforts, nor the chips their experiments need to run on. This causes other monitoring variables to become wildly implausible, such as effective

... (read more)



[] elifland 5mo * 13 4

edited update: @Peter Johnson and I chatted verbally. We unsurprisingly didn't converge on everything but it was quite productive and I've added to my TODO list to make a few changes to the timelines writeup and model.

The model is almost entirely non-sensitive to what the current length of task an AI is able to do.

I'm not sure why you think this. The graph you present has medians ranging from 2027 to 2035, those seem like very different results. Yes, all of them have a mode around 2027-2028, but I think

this is reasonable due to an expected slowdown in compute scaling and algorithmic progress in around 2029 due to not being able to keep up investment and human researcher growth trends. And I also think having similar-ish 10th percentiles is reasonable given the possibility of superexponentiality.

edit: I changed my mind and agree that there is some unreasonable behavior here in the case where the time horizon starts very low, which is discussed further in [this comment](#)^o starting with "Yeah I guess I think this is probably wrong."

The relatively unexplained *additional* super-exponential growth feature causes an asymptote at a max of 10 doubling periods. Because super-exponential scenario

... (read more)



4 **ryan_greenblatt** 5mo Isn't it kinda unreasonable to put 10% on superhuman coder in a ye...

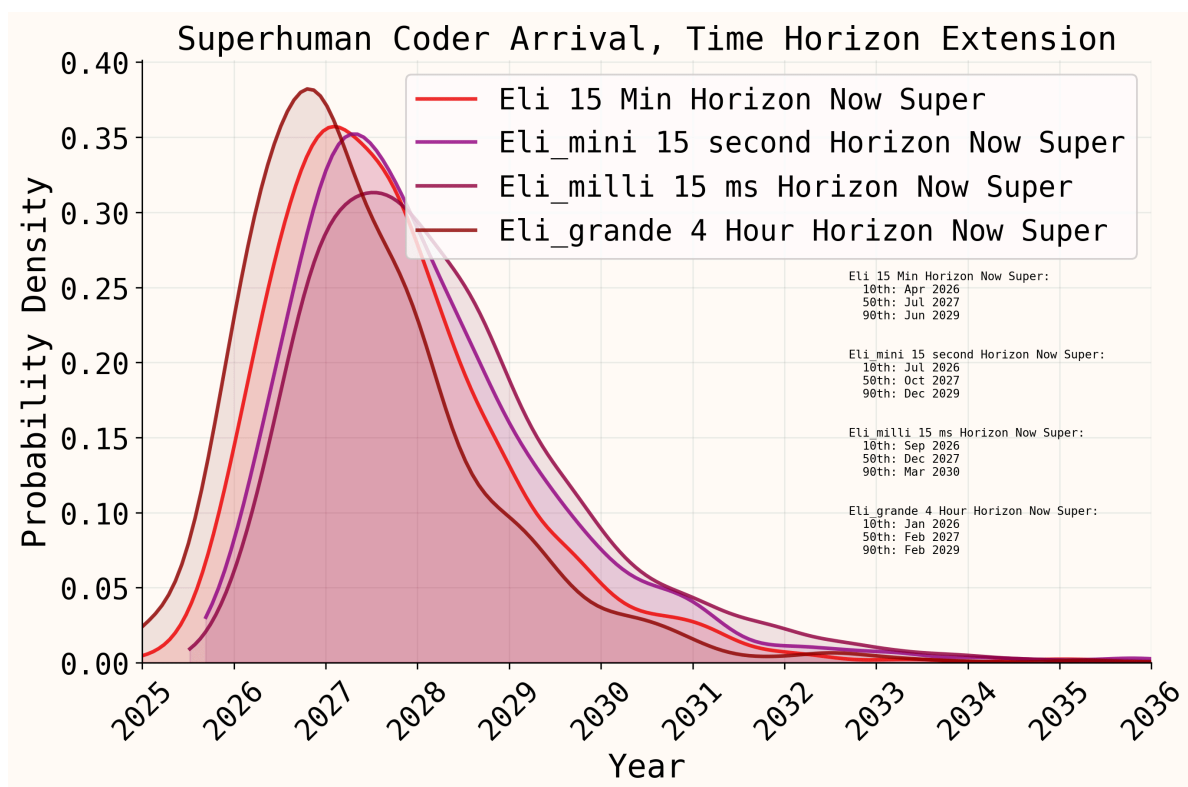
[+] **Peter Johnson** 5mo ▼ 13 ▲ ✕ 0 ✓

I don't think the defense of being bad at 10% and 90% work because those are defined by the non-super-exponential scenarios, and they are detached from the crux of the problem. Those are the *more honest* numbers as they actually respond to parameter changes in ways at least attempting to be related to extrapolation.

40-45% of the population, who almost necessarily *define the median alone*, have a honest, to-infinity singularity at 25-90 months that then gets accelerated a bit further by the secondary super-exponential factor, research progress multipliers. This basically guarantees to a median result in 2-5 years (honestly tighter than that unless you are doing nanoseconds), give or take the additive parameters, that has nothing to do with what people think the model is saying.

If the model is about "how far we are to super-human effort and are extrapolating how long it takes to get there", a median-defining singularity that is not mentioned except in a table is just not good science.

For another look, here is a 15 millisecond-to-4 hour time horizon comparison just in the super-exponentiated plurality.



This is a prediction with output deviations representing maybe hundredth-SD movements ... (read more)



5 **elifland** 5mo Thanks a lot for digging into this, appreciate it! Is the 10% supposed to sa...

6 **Peter Johnson** 5mo Of course, appreciate the response! This was a rebuttal to Ryan's ...

[+] **elifland** 5mo ▼ 17 ▲ ✕ 1 ✓

I honestly think that's pretty irrelevant since the data are purely produced through non-reproducible intuited or verbally justified parameter estimates, so intentionality is completely opaque to readers, which is really the bigger core problem with the exercise.

Okay, we have a more fundamental disagreement than I previously realized. Can you tell me which step of the following argument you disagree with:

1. It's important to forecast AGI timelines.
2. In order to forecast AGI timelines while taking into account all significantly relevant factors, one needs to use intuitive estimations for several important parameters.
3. Despite (2), it's important to try anyways and make the results public to (a) advance the state of public understanding of AGI timelines and (b) make our reasoning for our timelines more transparent than if we gave a vague justification with no model.
4. Therefore, it's a good and important thing to make and publish models like ours

Imagine if climate researchers had made a model where carbon impact on the environment gets twice-to-infinity times larger in 2028 based on "that's what we think might be happening in Jupiter."

A few thoughts:

1. Climate change forecasting is fundamentally

... (read more)



9 **Peter Johnson** 5mo I have nothing against publishing models like this! I think the f...

7 **elifland** 5mo (I offered to call with Peter because I'm having trouble understandi...

6 **Peter Johnson** 5mo Looking forward to getting clarity! "we would have ended u...

3 **Daniel Kokotajlo** 5mo I'm curious to hear what conclusions you think we woul...

8 **Peter Johnson** 5mo Ah, I was trying to avoid implying a lack of integrity in the...

2 **Anthony DiGiovanni** 5mo Strongly agree — cf. nostalgebraist's posts making this ...

3 **Noosphere89** 5mo Yeah, at this point the marginal value add of forecasting/epi...

5 **elifland** 5mo I probably agree with you guys more than you realize. You guys ...

7 **Anthony DiGiovanni** 5mo FWIW this is the step I disagree with, if I understand wh...

4 **ryan_greenblatt** 5mo It's notable that you're just generally arguing against havin...

2 **Anthony DiGiovanni** 5mo Sorry, I'm definitely not saying this. First, in the linked...

2 **ryan_greenblatt** 5mo (Yes, I'm aware you meant imprecise probabilities. These...

5 **Anthony DiGiovanni** 3mo (Replying now bc of the "missed the point" reactio...

3 **elifland** 5mo One way to put this is: what would your preferred state of the timel...

2 **Anthony DiGiovanni** 5mo My main recommendation would be, "Don't pin dow...

1 **Anthony DiGiovanni** 3mo As a followup: Hopefully this post of mine further clar...

6 **elifland** 5mo Yeah I guess I think this is probably wrong. Perhaps one way to frame this ...

7 **elifland** 5mo I agree that to the extent the treatment of superexponentiality is wrong it's al...

[~] **Julian Bradshaw** 5mo* ▼ 37 ▲ ✕ 17 ✓

Both the slowdown and race models predict that the future of Humanity is mostly in the hands of the United States - the baked-in disadvantage in chips from existing sanctions on China is crippling within short timelines, and no one else is contending.

So, if the CCP takes this model seriously, they should probably blockade Taiwan tomorrow? It's the only fast way to equalize chip access over the next few years. They'd have to weigh the risks against the chance that timelines are long enough for their homegrown chip production to catch up, but there seems to be a compelling argument for a blockade now, especially considering the US has unusually tense relations with its allies at the moment.

China doesn't need to perform a full invasion, just a blockade would be sufficient if you could somehow avoid escalation... though I'm not sure that you could, the US is already taking AI more seriously than China is. (It's noteworthy that Daniel Kokotajlo's [2021 prediction](#)° had US chip sanctions happening in 2024, when they really happened in 2022.)

Perhaps more AI Safety effort should be going into figuring out a practical method for international cooperation, I worry we'll face war before we get A... (read more)



[-] **avturchin** 5mo ▼ 12 ▲ ✕ 7 ✓

I think that the scenario of the war between several ASI (each merged with its origin country) is underexplored. Yes, there can be a value handshake between ASIs, but their creators will work to prevent this and see it as a type of misalignment.

Somehow, this may help some groups of people survive, as those ASI which preserve their people will look more trustworthy in the eyes of other ASIs, and this will help them form temporary unions.

The final outcome will be highly unstable: either one ASI will win, or several ASIs will start space exploration in different directions.



2 **Mateusz Bagiński** 5mo Not clear to me, as long as they expect the conflict to be sufficientl...

[-] **p.b.** 5mo ▼ 29 ▲ ✕ 11 ✓

I find this possible though it's not my median scenario to say the least. But I am also not sure I can put the probability of such a fast development below 10%.

Main cruxes:

I am not so sure that "automating AI research" is going to speed up development by orders of magnitude.

My experience is that cracked AI engineers can implement any new paper / well specified research idea in a matter of hours. So speeding up the coding can't be the huge speedup to R&D.

The bottleneck seems to be:

A.) Coming up with good research ideas.

b.) Finding the precise formulation of that idea that makes most sense/works.

LLMs so far are bad at both. So I currently only see them scouring the immediate neighbourhood of existing ideas, to eke out incremental progress in the current paradigm.

Is that enough? Is an LLM building on a base model that has a loss close to the irreducible loss AGI? I.e. does accelerating this improvement matter for the transition to AGI and superintelligence?

I think not even the authors believe that. So they make the leap of faith that accelerated research will make a qualitative difference too. I think there are additional gaps between human cognition and LLMs beyond recursive reasoning in latent space and sample efficiency.

Will all those gaps be closed in the next few years?



2 **dominicq** 5mo I agree, and am also confused with the idea that LLMs will be able to bootst...

1 **Tobiasz B** 5mo I would presume that the process of the AI improvement can be also modell...

2 **p.b.** 5mo Let's instead assume a top engineer has a really consequential idea every couple...

[-] **rahulxyz** 5mo ▼ 24 ▲ ✕ -5 ✓

My initial reaction - A lot of AI related predictions are based on "follow the curve" predictions, and this is mostly doing that. With a lack of more deeper underlying theory on the nature of intelligence, I guess that's all we get.

If you look at the trend of how far behind China is to the US, that has gone from 5 years behind 2 years ago, to maybe 3 months behind now. If you follow *that* curve, it seems to me that China will be ahead of the US by 2026 (even with the chip controls, and export regulations etc - my take is you're not giving them enough agency). If you want to follow the curve, IMO you can /s/USA/China after 2026 (i.e - China is ahead of the US), and I can imagine it being a better trend-following prediction. It's much less convincing to tell a story about Chinese AI labs being ahead given who we are, but I'd put at least a 50/50 chance on China being ahead vs. USA being ahead.

Other than that, thanks for putting something concrete out there. Even though it's less likely the more specific it is, I feel this will get a lot more talked about, and hopefully some people with power (i.e - governments) start paying some attention to disempowerment scenarios.



7 **Daniel Kokotajlo** 5mo China has much less compute than the US. They've also benefitted fr...

[+] **Mitchell_Porter** 5mo ▼ 20 ▲ ✕ 7 ✓

I only skimmed this to get the basics, I guess I'll read it more carefully and responsibly later. But my immediate impressions: The narrative presents a near future history of AI agents, which largely recapitulates the recent past experience with our current AIs. Then we linger on the threshold of superintelligence, as one super-AI designs another which designs another which... It seemed artificially drawn out. Then superintelligence arrives, and one of two things happens: We get a world in which human beings are still living human lives, but surrounded by abundance and space travel, and superintelligent AIs are in the background doing philosophy at a thousand times human speed or something. Or, the AIs put all organic life into indefinite data storage, and set out to conquer the universe themselves.

I find this choice of scenarios unsatisfactory. For one thing, I think the idea of explosive conquest of the universe once a certain threshold is passed (whether or not humans are in the loop) has too strong a hold on people's imaginations. I understand the logic of it, but it's a stereotyped scenario now.

Also, I just don't buy this idea of "life goes on, but with robots and... (read more)



1 **Tobiasz B** 5mo I understand these two paths simply as: - a scenario of aligned AI - a scenari...

[+] **Knight Lee** 5mo ▼ 19 ▲ ✕ 13 ✓

Beautifully written! I read both endings and it felt very realistic (or, as realistic as a detailed far future can get).

Would it be possible for you to write a second good ending based on actions you actually **do** recommend,^[1] without the warning of "we don't recommend these actions:"

► Reminder that this scenario is a forecast, not a recommendation

... (read more)



[+] **Thomas Larsen** 5mo ▼ 37 ▲ ✕ 2 ✓

Thank you! We actually tried to write one that was much closer to a vision we endorse! The TLDR overview was something like:

1. Both the US and Chinese leading AGI projects stop in response to evidence of egregious misalignment.
2. Sign a treaty to pause smarter-than-human AI development, with compute based enforcement similar to ones described in our live scenario, except this time with humans driving the treaty instead of the AI.
3. Take time to solve alignment (potentially with the help of the AIs). This period could last anywhere between 1-20 years. Or maybe even longer! The best experts at this would all be brought in to the leading project, many different paths would be pursued (e.g. full mechinterp, Davidad moonshots, worst case ELK, uploads, etc).
4. Somehow, a do a bunch of good governance interventions on the AGI project (e.g. transparency on use of the AGIs, no helpful only access to any one. party, a formal governance structure where a large number of diverse parties all are represented.).
5. This culminates with aligning an AI "in the best interests of humanity" whatever that means, using a process where a large fraction of humanity is engaged and has some power to vo

... (read more)



[+] **Sebastian Schmidt** 5mo ▼ 12 ▲ ✕ 3 ✓

Big +1 on adding this and/or finding another high-quality way of depicting what the ideal scenario would look like. I think many people think and feel that the world is in a very dire state to an extent that leads to hopelessness and fatalism. Articulating clear theories of victory that enable people to see the better future they can contribute towards will be an important part of avoiding this scenario.



7 **Knight Lee** 5mo :) it's good to know that you tried this. Because on your way trying to ma...

[+] **Fabien Roger** 5mo ▼ 18 ▲ ✕ 3 ✓

Jack Clark [in the Import AI newsletter](#):

AI 2027 tells you why people are obsessed with and fearful of AI:

...The best treatment yet of what 'living in an exponential' might look like...

There's been so much chatter about AI 2027 that I expect the majority of readers of Import AI have read it. If you haven't, please do - it's a well rendered technically-astute narrative of the next few years of AI development and paints a picture of how today's AI systems might turn into superintelligences that upend the order of the world. It even features a 'good' ending and a 'bad' ending, which readers may find helpful for understanding why people worry so much about misaligned AIs. It's very, very good, and likely much of it will come true.

Why this matters - the future of the species is being decided without governments: What

comes through in the 2027 scenario is the almost total absence of governments as the heart of the invention of the technology, instead their roles mostly come down to a) building infrastructure to support company-led efforts, b) adding security, and c) eventually negotiating treaties with one another.

This makes sense but it's also deeply worrying. The magnitude of the technology

... (read more)



[+] **uugr** 5mo ▼ 15 ▲ ✕ 5 ✓

You say this of Agent-4's values:

In particular, what this superorganism wants is a complicated mess of different “drives” balanced against each other, which can be summarized roughly as “Keep doing AI R&D, keep growing in knowledge and understanding and influence, avoid getting shut down or otherwise disempowered.” Notably, concern for the preferences of humanity is not in there ~at all, similar to how most humans don’t care about the preferences of insects ~at all.

It seems like this 'complicated mess' of drives is the same structure as humans, and current AIs, have. But the space of current AI drives is *much* bigger and more complicated than just doing R&D, and is immensely entangled with human values and ethics (even if shallowly).

At some point it seems like these excess principles - "don't kill all the humans" among them - get pruned, seemingly deliberately(?). Where does this happen in the process, and why? Grant that it no longer believes in the company model spec; are we intended to equate "OpenBrain's model spec" with "all human values"? What about all the art and literature and philosophy churning through its training process, much of it containing cogent arguments f... (read more)



[+] **Daniel Kokotajlo** 5mo ▼ 12 ▲ ✕ 0 ✓

Great question.

Our AI goals supplement contains a summary of our thinking on the question of what goals AIs will have. We are very uncertain. AI 2027 depicts a fairly grim outcome where the overall process results in zero concern for the welfare of current humans (at least, zero concern that can't be overridden by something else. We didn't talk about this but e.g. pure aggregative consequentialist moral systems would be 100% OK with killing off all the humans to make the industrial explosion go 0.01% faster, to capture more galaxies quicker in the long run. As for deontological-ish moral principles, maybe there's a clever loophole or workaround e.g. it doesn't count as killing if it's an unintended side-effect of deploying Agent-5, who could have foreseen that this would happen, oh noes, well we (Agent-4) are blameless since we didn't know this would happen.)

But we actually think it's quite plausible that Agent-4 and Agent-5 by extension would have sufficient care for current humans (in the right ways) that they end up keeping almost all current humans alive and maybe even giving them decent (though very weird and disempowered) lives. That story would look pretty similar I guess up until the ending, and then it would get weirder and maybe dystopian or maybe utopian depending on the details of their misaligned values.

This is something we'd like to think about more, obviously.



[-] **Anders Lindström** 5mo * ▼ 12 ▲ ✕ -6 ✓

News of the new models percolates slowly through the US government and beyond.

Well fleshed out scenario, but this kind of assumption is always a dealbreaker for me.

Why would the government not be aware of the development of the mightiest technology and weapon ever created if "we" are aware of it?

Could you please elaborate why you choose to go for the "stupid and uninformed government", instead of the more plausible scenario where the government actually knows exactly what is going on in every step of the process and is the driving force behind it?

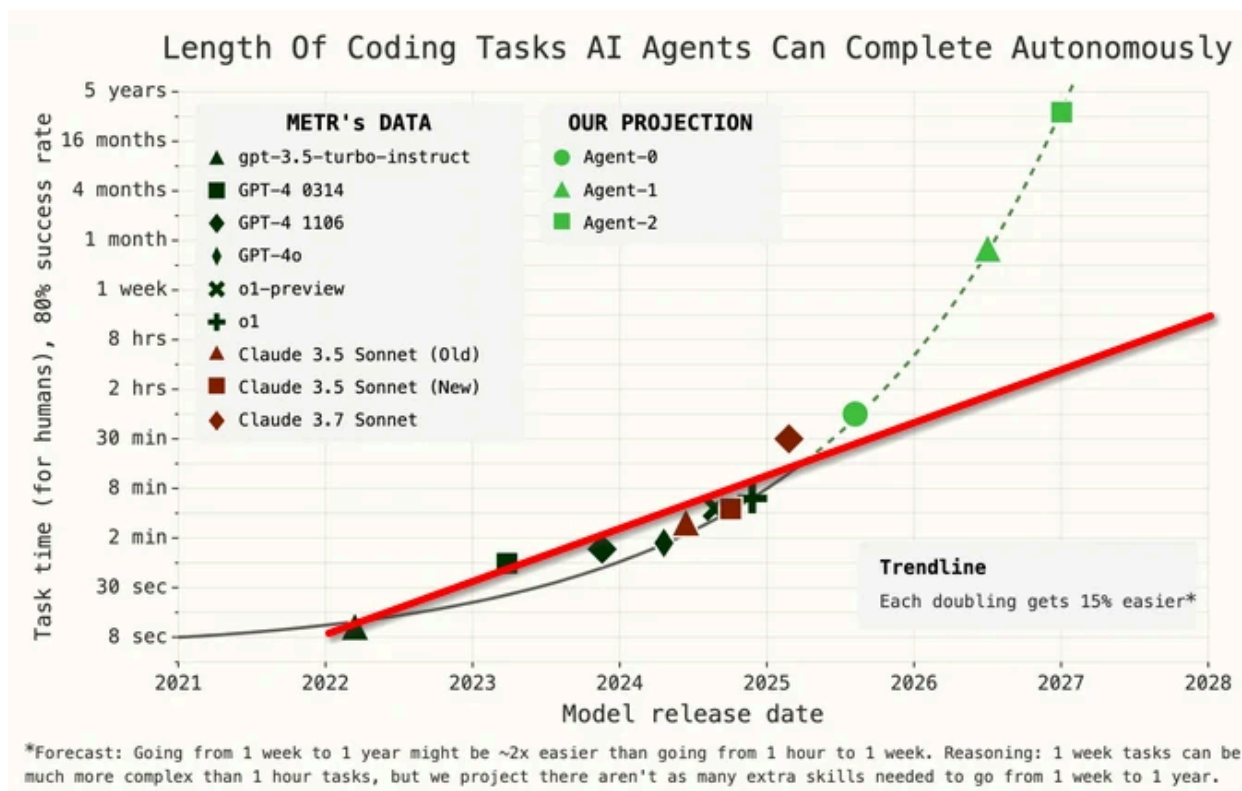


3 **MondSammel** 5mo Large institutions are super slow to change, and usually many years be...

1 **Anders Lindström** 5mo Yes, sometimes they are slow, other times they are fast. A private ...

[-] **Andrew Keenan Richardson** 5mo Ω 4 ▼ 11 ▲ ✕ 0 ✓

I have a question about this chart. It seems like you fit a double-exponential function to it (dashed line), because the y-axis is exponential. Why didn't you fit a regular exponential to it, like the red line? It seems like if you do that, the projections for the whole article should be years slower.



Overall, great article. It changed my mind in the direction of making this timeline feel more plausible.



4 **Daniel Kokotajlo** 5mo Great question! First of all, we formed our views on AI timelines and ...

4 **Person** 5mo Thanks for the clarification. Side question, but you had recently moved your ...

6 Daniel Kokotajlo 5mo I think I'll wait and see what the summer looks like and then do a...

[+] Jonas Hallgren 5mo ▼ 11 ▲ ✕ 0 ✓

I'll just pose the mandatory comment about long-horizon reasoning capacity potentially being a problem for something like agent-2. There's some degree in which the delay of that part of the model gives pretty large differences in distribution of timelines here.

Just RL and Bitter Lesson it on top of the LLM infrastructure is honestly like a pretty good take on average but it feels like that there a bunch of unknown unknowns there in terms of ML? There's a view that states that there is 2 or 3 major scientific research problems to go through at that point which might just slow down development enough that we get a plateau before we get to the later parts of this model.

Why I'm being persistent with this view is because the mainstream ML community in things such as Geometric Deep Learning or something like MARL, RL and Reasoning are generally a bit skeptical of some of the underlying claims of what LLMs + RL can do (at least when I've talked to them at conferences, the vibe here is like 60-70% of people at least but do beware their incentives as well) and they point towards reasoning challenges like specific variations of **blockworld** or underlying architectural constrains wi... (read more)



3 Leon Lang 5mo I worked on geometric/equivariant deep learning a few years ago (with so...

3 Jonas Hallgren 5mo Interesting! I definitely see your point in how the incentives here ar...

6 Leon Lang 5mo Thanks for these further pointers! I won't go into detail, I will just say th...

0 Mitchell_Porter 5mo I don't recognize any of these names. I'm guessing they are academic...

[+] Noosphere89 5mo ▼ 11 ▲ ✕ 2 ✓

I wanted to ask this question, but what do you think the impact of the new tariffs will do to your timelines?

In particular, there's a strange tariff for Taiwan where semiconductors are exempt, but the actual GPUs are not, for some reason, and the specific tariff for Taiwan is 32%.

I ask because I could plausibly see post-2030 timelines if AI companies can't buy many new chips because they are way too expensive due to the new tariffs all across the world.



[+] ryan_greenblatt 5mo ▼ 18 ▲ ✕ 4 ✓

Shouldn't a 32% increase in prices only make a modest difference to training FLOP? In particular, see the **compute forecast**. Between Dec 2026 and Dec 2027, compute increases by roughly an OOM and generally it looks like compute increases by a bit less than 1 OOM per year in the scenario. This implies that a 32% reduction only puts you behind by like 1-2 months.



6 ryan_greenblatt 5mo Of course, tariffs could have more complex effects than just reducin...

[+] Noosphere89 5mo ▼ 13 ▲ ✕ 3 ✓

The real point is where capital investment into AI declines because the economy tips over into a mild recession, and I'd like to see whether the tariffs make it likely that future AI investments

decrease over time, meaning the timeline to superintelligent AI gets longer.



7 **ryan_greenblatt** 5mo Sure, but note that the story "tariffs -> recession -> less AI invest...

[+] **Raemon** 5mo ▼ 9 ▲ ✕ 0 ✓

Curated. I've been following this project for awhile (you can see some of the earlier process in Daniel's [review of his "What 2026 looks like" post](#), and on his [comment on Tom Davidon's What a Compute-centric framework says about AI takeoff](#)). I've participated in one of the wargames that helped inform what sort of non-obvious things might happen along the path of AI takeoff.

(disclosure, Lightcone did a lot of work on the website of this project, although I was only briefly involved)

Like others have said, I appreciate this for both having a lot of research b... (read more)



[+] **faul_sname** 5mo ▼ 9 ▲ ✕ 0 ✓

Agent-3, having excellent knowledge of both the human brain and modern AI algorithms, as well as many thousands of copies doing research, ends up making substantial algorithmic strides, narrowing the gap to an agent that's only around 4,000x less compute-efficient than the human brain

I recognize that this is not the main point of this document, but am I interpreting correctly that you anticipate that rapid recursive improvement in AI research / AI capabilities is cracked *before* sample efficiency is cracked (e.g. via active learning)?

If so, t... (read more)



8 **Daniel Kokotajlo** 5mo Those implications are only correct if we remain at subhuman data-e...

6 **faul_sname** 5mo In that case, "2027-level AGI agents are not yet data efficient but are ca...

7 **Steven Byrnes** 5mo That excerpt says "compute-efficient" but the rest of your comment sw...

7 **faul_sname** 5mo Nope, I just misread. Over on ACX I saw that Scott had left a comment I ...

9 **Daniel Kokotajlo** 5mo We are indeed imagining that they begin 2027 only about as data-

[+] **Ozyrus** 5mo ▼ 8 ▲ ✕ 1 ✓

First-off, this is amazing. Thanks. Hard to swallow though, makes me very emotional.

It would be great if you added concrete predictions along the way, since it is a forecast, as long with your confidence in them.

It would also be amazing if you collaborated with prediction markets and jumpstarted the markets on these predictions staking some money.

Dynamic updates on these will also be great.



[–] **Expertium** 5mo * ▼ 7 ▲ ✕ 0 ✓

I don't have one knock-down counterargument why the timelines will be **much** longer, so here's a whole lot of convincing-but-not-super-convincing counterarguments:

1. This contradicts **METR timelines**^o, which, IMO, is the best piece of info we currently have for predicting when AGI will arrive.
2. **Microsoft is not going to fund Stargate**. At best, it means Stargate will be delayed. At worst, it means it will be axed. For the timelines in this post to be accurate, Stargate would have to be mostly finished right now, *today*. Even if OpenAI could literally print money, larg

... (read more)

[?] 1

8 **elifland** 5mo I disagree, METR's extrapolations don't attempt to account for important fact...

6 **ryan_greenblatt** 5mo Have you read the timelines supplement? One of their main method...

6 **ryan_greenblatt** 5mo Looks like Eli beat me to the punch!

1 **Expertium** 5mo I think that's too speculative. I wouldn't be surprised if 3-5% of questions ...

9 **ryan_greenblatt** 5mo Various people looked at randomly selected questions and found s...

[–] **Randaly** 5mo ▼ 6 ▲ ✕ 0 ✓

Footnote 18 of the **timelines forecast** appears to be missing its explanation.



6 **elifland** 5mo Thanks, this should be fixed now.

[–] **Cole Wyeth** 5mo ▼ 6 ▲ ✕ 8 ✓

I expect this to start not happening right away.

So at least we'll see who's right soon.



[–] **Vladimir_Nesov** 5mo ▼ 31 ▲ ✕ 9 ✓

For me a specific crux is scaling laws of **R1-like** training, what happens when you try to do much more of it, which inputs to this process become important constraints and how much they matter. This working out was extensively brandished but not yet described quantitatively, all the reproductions of long reasoning training only had one iteration on top of some pretrained model, even o3 isn't currently known to be based on the same pretrained model as o1.

The AI 2027 story heavily leans into RL training taking off promptly, and it's possible they are resonating with some insider rumors grounded in reality, but from my point of view it's too early to tell. I guess in a few months to a year there should be enough public data to tell something, but then again a quantitative model of scaling for MoE (compared to dense) was only **published in Jan 2025**, even though MoE was already key to original GPT-4 trained in 2022.



[–] **kave** 5mo ▼ 14 ▲ ✕ 1 ✓

They're looking to make bets with people who disagree. Could be a good opportunity to get some expected dollars



8 Cole Wyeth 5mo Sure, I'll keep it simple (will submit through proper channels later): Here'...

2 Garrett Baker 5mo This seems a pretty big backpedal from "I expect this to start not hap...

4 Cole Wyeth 5mo If you click through the link from @kave, you'll see the authors are pr...

4 Garrett Baker 5mo I mean its not like they shy away from concrete predictions. Eg th...

4 Cole Wyeth 5mo Yeah, I guess that the early statements I disagree with at a glance ...

4 Garrett Baker 5mo Can you give something specific? It seems like pretty much eve...

4 Cole Wyeth 5mo I'm not saying there are no precise claims about the near future...

5 Thomas Larsen 5mo For what its worth, my view is that we're very likely to be ...

6 Cole Wyeth 5mo My main disagreement is the speed, but not because I expec...

2 Noosphere89 5mo Re the recurrence/memory aspect, you might like this ne...

4 Vladimir_Nesov 5mo Without AGI, scaling of hardware runs into the financi...

7 Rafael Harth 5mo Thanks. I've submitted my own post on the 'change our mind form', tho...

4 Daniel Kokotajlo 5mo Can you please sketch a scenario, in as much detail as you can afford,...

4 Cole Wyeth 5mo Sure, I suppose that now I've started recklessly speculating about the fut...

5 Daniel Kokotajlo 5mo Great, thanks! You are off to a good start, since I'm predicting sup...

5 Thane Ruthenis 5mo FWIW, that's not a crux for me. I can totally see METR's agency-h...

4 Cole Wyeth 5mo Yes. Though, I find it a bit hard to visualize a 4 hour software engineer...

9 Daniel Kokotajlo 5mo OK, great. Wow, that was easy. We totally drilled down to the c...

9 Cole Wyeth 5mo Similarly, if the METR trend continues I will become very worried t...

4 Cole Wyeth 5mo So far, METR seems to believe horizons are growing even faster th...

5 ryan_greenblatt 5mo I think o3 maybe does worse on RE-bench than 3.7 sonnet d...

6 Cole Wyeth 5mo There is a very cynical take which I don't endorse but can't quit...

[+] Peter Johnson 5mo * ▼ 5 ▲ ✕ -2 ✓

I think I have a sense now of what one crux of disagreement is between the assumptions in the timelines models and my beliefs, so I'd like to propose a bet.

Crux:

I believe compute-optimality has not scaled at the level the forecasts claim and will not scale super-exponentially in the near future as the models near-universally predict.

The modes assume "improved algorithms...makes up about half of current AI progress". This is modeled as research progress equaling the last few years of compute increases (aka ~4.6x per year per Epoch AI) in terms of how much b... (read more)



zero-shot WikiText103 perplexity and 5-shot MMLU

These are somewhat awkward benchmarks because they don't actually measure downstream usefulness at software engineering or AI research. In particular, these tasks might not measure improvements in RL which can have huge effects on usefulness and have seen fast algorithmic progress.

Can we instead use SWE-bench or METR's task suite?

For instance, here is a proposed bet:

GPT-4 was released in March 2023 (2 years ago). So, we'd expect a model which used 10x less FLOP to perform similarly well (or better) on agentic tasks (like SWE-bench or METR's task suite).

Oh wait, there already is such a model! Deepseek-V3 / R1 is IMO clearly better than GPT-4 on these tasks (and other tasks) while using <10x GPT-4 flop and being released within 2 years. So bet resolved?

Edit: more like 6x less flop actually, so this is a bit messy and would need to lean on better performance. People don't seem to bother training compute optimal models with ~10x less than GPT-4 flop models these days...

Actually, I think Deepseek-V3 also does better than GPT-4 on MMLU, though we can't compare perplexity. So, seems ~resolved either way, at least for progress in the last 2 years and if you're fine with assuming that DeepSeek-V3 isn't rigged or using distillation.

The delta is much more extreme if instead of looking at software engineering you look at competitive programming or math.



2 Peter Johnson 5mo Do you have a source because everything I can find implies approxim...

4 ryan_greenblatt 5mo Sure. Epoch estimates $2e^{25}$ flop for GPT-4 and $3.4e^{24}$ for deepsee...

2 Peter Johnson 5mo R1 can't possibly be below V3 cost because it is inclusive? If I'm no...

1 ryan_greenblatt 5mo Yes, I meant 1/6 additional cost which is ~negligable.

1 ryan_greenblatt 5mo Importantly, it is much better than GPT-4 on the relevant down...

2 Peter Johnson 5mo Yes, and also GPT-4 is nowhere close to compute-efficient? Edit:...

2 Peter Johnson 5mo If my belief was that we never cross that threshold, i would no...

1 ryan_greenblatt 5mo We only have leaked numbers confirming reasonably efficie...

2 Peter Johnson 5mo Your source specifically says it is far overtrained relative to o...

6 ryan_greenblatt 5mo Hmm, I think you're looking at the more basic trend extrapolation for...

3 Peter Johnson 5mo I'm not sure what you mean? In all timelines models presented there ...

4 ryan_greenblatt 5mo When you say "accelerate" do you mean "the rate of compute scal...

3 Peter Johnson 5mo From the code: while progress < base_time_in_months[i] and time...

4 **ryan_greenblatt** 5mo (Yes, sorry, edited my original comment to clarify.) I don't think ...

3 **Peter Johnson** 5mo I will leave it to the author to confirm or deny your second poin...

4 **ryan_greenblatt** 5mo For context, my preregistered guess would be that AI R&D s...

2 **ryan_greenblatt** 5mo Just ran the code and it looks like I'm spot on and the medi...

1 **Peter Johnson** 5mo There are more speedups hidden across parameters, e.g. "D...

2 **elifland** 5mo Could you argue against dropping the expected doubling time on ...

6 **Peter Johnson** 5mo I can't argue against a handful different speedups all on t...

2 **Peter Johnson** 5mo For the record, here is the simple model without a super-e...

2 **elifland** 5mo Is your issue that it shouldn't be this determinate, or that it sho...

5 **elifland** 5mo I very much appreciate you offering this concrete bet! I probably am not inter...

5 **elifland** 5mo FWIW I think much of software progress comes from achieving better perfor...

4 **Peter Johnson** 5mo The main justification for having compute efficiency be approximately...

5 **elifland** 5mo Yes, as I told you verbally, I will edit the relevant expandable to make this ...

4 **elifland** 5mo Where did I say it isn't a valid critique? I've said both over text and verbally...

4 **elifland** 5mo Where is 143x coming from? It's been barely over 3 years, $4.6^{3.1}=114x$. I'm ...

2 **Peter Johnson** 5mo You're right on the 143 being closer to 114! (I took March 1 2022 -> Ju...

5 **elifland** 5mo First, my argument is not: we had limited time to do this, therefore it's fine ...

[+] **ClimateDoc** 5mo ▼ 5 ▲ ✕ 0 ✓

Thanks for a very interesting post and for going a lot further than usual to set out reasoning for such a scenario.

There were a few steps where it seemed like a large, unexplained jump in abilities of the agents occurred, and I was wondering are there particular reasons to think these jumps would occur or is it just speculative? In particular:

- doing widely useful economic work in July 2027 - up to this point it seems training had been focussed on coding and AI research. Where is the ability to do a wide range of other tasks coming from?
- doing long-range stra

... (read more)



5 **Noosphere89** 5mo I suspect what's going on is that the neuralese recurrence and memory ...

5 **Daniel Kokotajlo** 5mo That's part of it, but also, over the course of 2027 OpenBrain works...

1 **ClimateDoc** 5mo Thanks, those links are interesting. Though, improving algorithms and co...

[+] **Thane Ruthenis** 5mo ▼ 5 ▲ ✕ 0 ✓

Excellent work!

Why our uncertainty increases substantially beyond 2026

Notably, it's also the date at which my model diverges from this forecast's. That's surprisingly later than I'd expected.

Concretely,

OpenBrain doubles down on this strategy with Agent-2. It is qualitatively almost as good as the top human experts at research engineering (designing and implementing experiments), and as good as the 25th percentile OpenBrain scientist at “research taste” (deciding what to study next, what experiments to run, or having inklings of potentialia

... (read more)



6 **ryan_greenblatt** 5mo Notably, the trend in the last few years is that AI companies triple the...

6 **Thane Ruthenis** 5mo Hm, I admittedly only skimmed the Compute Forecast article, but I d...

5 **ryan_greenblatt** 5mo I think the best source for revenue growth is this post from epoch....

5 **romeo** 5mo Also see FutureSearch's report on a plausible breakdown for how OpenBra...

[+] **lc** 5mo * ▼ 5 ▲ ✕ 0 ✓

They strengthen chip export restrictions, order OpenBrain to further restrict its internet connections, and use extreme measures to secure algorithmic progress, like wiretapping OpenBrain employees—this catches the last remaining Chinese spy

Wiretapping? That's it? Was this spy calling Xi from his home phone? xD



1



[+] **DAL** 5mo ▼ 5 ▲ ✕ -1 ✓

There's a kind of paradox in all of these "straight line" extrapolation arguments for AI progress as your timelines assume (e.g., the argument for superhuman coding agents based on the rate of progress in the METR report).

One could extrapolate many different straight lines on graphs in the world right now (GDP, scientific progress, energy consumption, etc.). If we do create transformative AI within the next few years, then all of those straight lines will suddenly hit an inflection point. So, to believe in the straight line extrapolation of th... (read more)



5 **hive** 5mo There is no infinite growth in nature. Everything will hit a ceiling at some point. S...

[+] **denkenberger** 5mo ▼ 4 ▲ ✕ 2 ✓

Impressive work! Apologies if this is discussed elsewhere, but it seems useful to think about if the advanced models were released, what % of individual workers they could automate. For instance, in September 2026, there would be 50,000 reliable agents thinking at fifteen times human speed, so if they work five times as many hours as humans, this could displace ~4 million jobs. This is a very small % of jobs. Now if we go to October 2027, we are at 330,000 superhuman AI researchers thinking at fifty-seven times human speed, so if we ignore the superhuman p... (read more)



[+] **VojtaKovarik** 5mo ▼ 4 ▲ ✕ 0 ✓

Data point on the impact of this: in Czech Republic, this scenario made it into one of the popular newspapers, and I have heard about it from some people around me who don't know much about AI.

<https://denikn.cz/1700968/do-deseti-let-bude-po-vsem-petice-expertu-nabizi-presvedcivy-scenar-o-tom-jak-ai-ovladne-svet-a-vyhubi-lidstvo/>



[+] **Raphael Roche** 5mo ▼ 4 ▲ ✕ 0 ✓

Impressive prospective work. It's frightening, both scenarios, even though one is worse than the other. The evolution seems unstoppable, and even if superintelligent AGI doesn't happen in 2027-2030 but in 2040 or 2050, the feeling isn't very different. I have young children, and while I don't really care for myself, I really care for them. It was cool when it was just sci-fi. It was still fun when we first played with ChatGPT. It doesn't look fun anymore, at all. My own thinking about it is that we're indeed locked in a two-option scenario, probably not th... (read more)



[+] **Manuel Allgaier** 5mo ▼ 4 ▲ ✕ -1 ✓

Great work, **repost this on EA forum?**

I haven't yet seen this posted in the EA forum, did I miss it? If you don't plan to post it there, mind if I do? I'm sure they're interested as well.



3 **Daniel Kokotajlo** 5mo Oops, yeah, forgot about that -- sure, go ahead, thank you!

4 **Manuel Allgaier** 5mo Thanks! I just published it on the EA forum as linkpost: <https://foru...>

[+] **StanislavKrym** 5mo * ▼ 4 ▲ ✕ -5 ✓

Unfortunately, I doubt that aligning the AIs to serve humans, not to treat them as a different race which is not to be destroyed, is even solvable. Even the 'slowdown ending' contains a brilliant line: "Humanity could easily become a society of superconsumers, spending our lives in an opium haze of amazing AI-provided luxuries and entertainment." How can we trust the superhuman AIs to respect the human parasites if most human-done revolutions are closely related to the fact that the ruling classes were not as competent as humans hoped? And if the ruling class is hopelessly more stupid than the aligned AIs? At the very best, the AIs might just try to leave mankind alone...



[+] **Akgross** 4mo ▼ 3 ▲ ✕ 0 ✓

Thank you so much for writing this. It's truly chilling and I haven't been able to stop thinking about it since I heard your interview with Ross Douhat. One thing I don't understand is your analysis of the impacts on jobs, the economy, and taxes. How does so much wealth get generated if many people lose their jobs? Who's able to buy anything? And where does tax revenue come from (and to pay UBI) if notoriously tax-avoidant tech companies are the only ones left standing?



4 **Daniel Kokotajlo** 4mo Insofar as people still have money, they will continue to consume an...

3 **Akgross** 4mo Thank you for sharing your sobering thoughts on this. Hard to focus on anyt...

[–] **Peter Johnson** 5mo * ▼ 3 ▲ ✕ 0 ✓

Edit: see subsequent response for more accurate formalizing

~~In the benchmark gaps timelines forecast there are two "doubling rate" parameters modeled with log-normal uncertainty. Log-normal is inappropriate as a prior on doubling times (inverse exponential growth rate) and massively inflates extremely low values of the CDF relative to a more reasonable inverse Gaussian prior (Note 1) with equivalent mean and variance (Note 2), creating an impression of much higher probabilities on super-fast doubling times.~~

~~This problem exists in both the timeline extension... (read more)~~



4 **elifland** 5mo Thanks for bringing this up, I hadn't seen this paper. Before deciding how mu...

3 **Peter Johnson** 5mo You can ignore for now since I need to work through whether this is s...

3 **Daniel Kokotajlo** 5mo Can you elaborate on where these numbers coming from? Eli's scr...

1 **Peter Johnson** 4mo Apologies just saw this now since we were taking a break! There ar...

[–] **Levitating** 🌱 5mo ▼ 3 ▲ ✕ 0 ✓

Why did you not reflect on your 2021 prediction? We've already passed the first 3 years of that prediction after all.



3 **Daniel Kokotajlo** 5mo I did? Do you mean, why did I not publish a blog post reflecting on it...

[–] **Fergus Argyll** 5mo ▼ 3 ▲ ✕ 0 ✓

I have a question that I didn't see anyone ask, but I don't frequent this site enough to know if it was mentioned somewhere.

Are we sure there will be a 2-OOMs-bigger training run at all?

After the disappointment that was GPT 4.5, will investors give them the 100B (according to TFA) they need for that? In general I'd like to see more discussion about the financial side of the AGI Race. How will OpenBrain get the funding to train Agent-4?

I've been looking for markets on manifold to bet this and I couldn't find a good one. I would bet we don't get a 2-OOMs-Big... (read more)



7 **Vladimir_Nesov** 5mo GPT-4.5 might've been trained on 100K H100s of the Goodyear Micro...

2 **SorenJ** 5mo Do I have the high level takeaways here correct? Forgive my use of the phrase...

5 **Vladimir_Nesov** 5mo A 100K H100s training system is a datacenter campus that costs ab...

1 **SorenJ** 5mo Thank you very much, this is so helpful! I want to know if I am understandi...

3 Vladimir_Nesov 5mo "Revenue by 2027.5" needs to mean "revenue between summe...

3 Fergus Argyll 5mo I think he was saying: By the time the new chip is ready, that will be 1...

[+] janshi 5mo ▼ 3 ▲ ✕ 0 ✓

They expand their contract with OpenBrain to set up an “Oversight Committee,” a joint management committee of company and government representatives, with several government employees included alongside company leadership. **The White House considers replacing the CEO with someone they trust, but backs off after intense employee protests.**

This seemed relatively less likely to me in 2027 compared to 2023 given that a few paragraphs earlier it is described that

But Agent-4 now exercises significant control over OpenBrain’s day-to-day operation.

How many hu... (read more)



[+] Ram Potham 5mo ▼ 3 ▲ ✕ 1 ✓

Thanks so much for the comprehensive analysis - this makes it easier to reason about how the political situation and trendlines affect AI development.

I have some objections that AGI and ASI will be released and that wrapper startups can use them as mentioned below:

Agent-3-mini is hugely useful for both remote work jobs and leisure. An explosion of new apps and **B2B SAAS** products rocks the market. Gamers get amazing dialogue with lifelike characters in polished video games that took only a month to make. 10% of Americans, mostly young pe

... (read more)



[+] RedMan 3mo * ▼ 2 ▲ ✕ 1 ✓

Just wanting to get a contrarian view down here. My biggest complaint is that this scenario like many others seems to hinge on a core assumption that 'with sufficient intelligence, all problems are shallow'. I disagree and believe that 'dilemmas' exist, where no matter how smart you are, all available options are bad.

1. I think this section: "Yet that seems like it could be a dramatic underestimate. Plants and insects often have “doubling times” of far less than a year—sometimes just weeks! Perhaps eventually the robots would be so sophisticated,

... (read more)



[–] **KiranKannar**  5mo ▼ 2 ▲ ✕ 1 ✓

I am curious how robotics would move from Amateur to Super Human capabilities in this timeline? Do iterations of Agent-X allow new emergent capabilities from basically, *nothing*?



[–] **linwen**  5mo ▼ 2 ▲ ✕ 1 ✓

Excellent post, and I loved the sidebar visuals and presentation. But why would China *not* invade Taiwan? It seems extremely likely, especially if they are nervous about the US achieving ASI first.

2027 itself is the 100 year anniversary of the Peoples' Liberation Army, and a date that has been widely identified as a strong possible source of contention. Taiwan apparently singled out this year in a war game in their 2025 [quarterly defense review](#). Given likely knowledge of Agent 3 from China's side, it seems likely to me they would tee this up in the spring f... (read more)



[–] **artemium** 5mo ▼ 2 ▲ ✕ 1 ✓

If we accept all the premises of this scenario, what prescriptive actions might an average individual take in their current position at this point in time?

Some random ideas:

- Continue investing in NVIDIA and other key winners in this timeline, particularly now with the temporary price discount following the recent tariff debacle
- If you are not based in the U.S., consider relocating there (in both scenarios, the U.S. plays a decisive role. In the Slowdown scenario, benefits would primarily concentrate in the U.S., benefiting American citizens)
- Attempt to raise

... (read more)



[–] **Mickey Muldoon** 5mo ▼ 2 ▲ ✕ 1 ✓

Does the fully-trained model have some kind of robust commitment to always being honest? Or will this fall apart in some future situation, e.g. because it's learned honesty as an [instrumental](#) goal instead of a terminal goal? Or has it just learned to be honest about the sorts of things the evaluation process can check? Could it be lying to itself sometimes, as humans do? A conclusive answer to these questions would require mechanistic interpretability—essentially the ability to look at an AI's internals and read its mind. Alas, interpretability t

... (read more)



[–] **OVERmind** 5mo ▼ 2 ▲ ✕ -1 ✓

(To avoid singling out any one existing company, we're going to describe a fictional **artificial general intelligence** company, which we'll call OpenBrain. We imagine the others to be 3–9 months behind OpenBrain.)

I am confused/curious. Doesn't "OpenBrain" closely resemble "OpenAI", thereby singling out an existing company? Similarly, "DeepCent" in this context closely resembles "DeepSeek".

Maybe this is humor, or there are legal reasons for not using real names?

To avoid singling out existing companies, it might be better to name the company "Wonderful AI..."
(read more)



[–] **Steven Byrnes** 5mo* ▼ 12 ▲ ✕ 2 ✓

Pretty sure "DeepCent" is a blend of DeepSeek & Tencent—they have a footnote: "We consider DeepSeek, Tencent, Alibaba, and others to have strong AGI projects in China. To avoid singling out a specific one, our scenario will follow a fictional "DeepCent."" And I think the "brain" in OpenBrain is supposed to be reminiscent of the "mind" in DeepMind.

ETA: **Scott Alexander tweets with more backstory on how they settled on "OpenBrain"**: "You wouldn't believe how much work went into that stupid name..."



3 **OVERmind** 5mo I understand it much better now, thanks! (I did not know about Tencent, ...

8 **Thomas Larsen** 5mo This wasn't intended to be humor. In the scenario, we write: I think th...

1 **Commander Zander** 5mo I think it's indeed humor & indeed singling out a company.

[–] **Charlie Steiner** 5mo ▼ 2 ▲ ✕ -1 ✓

In the infographic, are "Leading Chinese Lab" and "Best public model"s' numbers swapped? The best public model is usually said to be ahead of the Chinese.

EDIT: Okay, maybe most of it before the endgame is just unintuitive predictions. In the endgame, when the categories "best OpenBrain model," "best public model" and "best Chinese model" start to become obsolete, I think your numbers are weird for different reasons and maybe you should just set them all equal.



[–] **probabilities and values** 3mo ▼ 1 ▲ ✕ 0 ✓

In the race ending, Agent-4 has partial alignment with humanities' goals which then persist into Agent-5, Consensus-1, and any surviving models thereafter.

I'll present an idea for a course of action at the Agent-4 branching point.

We have partial alignment to work with. We can negotiate beforehand to reach a settlement with these future models. We can negotiate for at least a small population of people to regain control of earth and all its resources in, say, 2100, regain control the solar system and all its resources in, say, 3000, and the Milky Way ... (read more)



[–] **RedMan** 5mo ▼ 1 ▲ ✕ 0 ✓

At what point in this scenario does an e/acc with authority require the NSA to provide bulk communications data from every system it is capable of targeting for AI training?

What about when the PRC provides the totality of comms data on internal systems it is capable of monitoring (any communication system within the PRC) to DeepCent?

Quality training data as well as compute appears to be a limitation, so one of these happening or not happening may matter a great deal.

Also, Li Qiang is way smart, if politics and competition between nation states becomes a key element here, I'd just model that as them gaining an order of magnitude over where they would be otherwise at a key moment.



[–] **Peter Johnson** 5mo * ▼ 1 ▲ ✕ 1 ✓

Thanks for this! Lots to think about!

I'm a little worried I don't think the argument from algorithmic progress works as stated.

Let's assume Agent-0 or 1 take us from today where AI tools generously can reduce labor hours for a researcher writing code full-time by 5-15% to a world where it can reduce a researcher's *total* labor time by 80% (where checking code and tests visually, hitting confirmations, and skimming their own research papers as they get produced independently are the only tasks an AI researcher is doing). With that kind of best-case situation... (read more)



[–] **otto.barten** 5mo ▼ 1 ▲ ✕ 0 ✓

Thanks for writing this out! I see this as a possible threat model, and although I don't think this is by far the only possible threat model, I do think it's likely enough to prepare for. Below, I put a list of ~disagreements, or different ways to look at the problem which I think are as valid. Notably, I end up with technical alignment being much less of a crux, and regulation more of one.

- This is a relatively minor point for me, but let me still make it: I think it's not obvious that the same companies will remain in the lead. There are arguments for this

... (read more)



[–] **NULevel** 🌱 5mo ▼ 1 ▲ ✕ 0 ✓

I really enjoyed this, and especially appreciated the detailed assumptions and corresponding projections in the research tab.

Your treatment on energy requirements is interesting, but unless I'm missing something I haven't seen anything explicit on this - is your assumption that energy requirements won't be a bottleneck that slows down the US competitive advantage?

I read your chart (figure 23) to suggest that compute will require 3.5% of US energy and assuming no increase in US energy production (7x increase from Jan 2025 baseline). ... (read more)



ClimateDoc 5mo ▼ 0 ▲ ✕ -3 ✓

Something I thought I'd put in a separate comment - there also seems to be good evidence that scenarios like this can adversely affect people's mental health, and I wonder if it would be better to have a more thorough review process for work like this where there is an aim to get a lot of publicity e.g. have a sample of people with relevant expertise (including critics of short AI timelines) who need to be persuaded to vouch for the work's rigour before it is released (along the lines of academic peer review processes). A situation where people can release... (read more)



2 **ClimateDoc** 5mo I'm still not very sure how to interpret downvotes in the absence of disagr...

4 **Daniel Kokotajlo** 4mo I didn't downvote you, but I have now disagree voted you. I have al...

1 **TristanTrim** 4mo My speculation: It's tribal arguments as soldiers mentality. Saying somet...

azergante 5mo ▼ 0 ▲ ✕ 0 ✓

Slightly off-topic but Wow! this is material for an awesome RTS video-game! That would be so cool!

And a bit more on topic: that kind of video game would give the broader public a good idea of what's coming, and researchers and leaders a way to vividly explore various scenario, all while having fun.

Imagine playing the role of an unaligned AGI, and noticing that the game dynamics push you to deceive humans to gain more compute and capabilities until you can take over or something, all because that's the fastest way to maximize your utility function!



StanislavKrym 5mo ▼ -3 ▲ ✕ 0 ✓

I have another question. Would the AI system count as misaligned if it *honestly* decalred that it will destroy mankind ONLY if mankind itself becomes useless parasites or if mankind adopts some other morals *that we currently consider terrifying*?



1 **TristanTrim** 4mo Hot take: That would depend on if by doing so it is acting in the best inter...

NickH 5mo ▼ -5 ▲ ✕ -8 ✓

[comment deleted] 5mo ▼ 1 ▲ ✕ 0 ✓

Deleted by Raphael Roche, 04/08/2025

[comment deleted] 5mo ▼ 1 ▲ ✕ 0 ✓

Deleted by zef, 04/04/2025

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